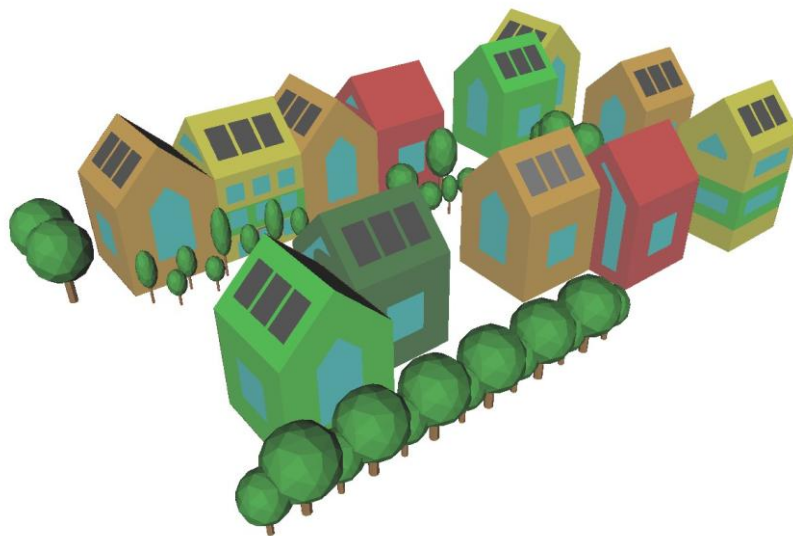




Beta 8

UML diagrams



Last updated on 28 August 2026





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The latest version of this document can be downloaded from:

https://github.com/tudelft3d/Energy_ADE/tree/3.0beta8/documentation

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Driving Urban
Transitions



Co-funded by
the European Union



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The author would like to thank the following people for their suggestions and fruitful discussions that have helped design the current version of the Energy ADE conceptual data model:

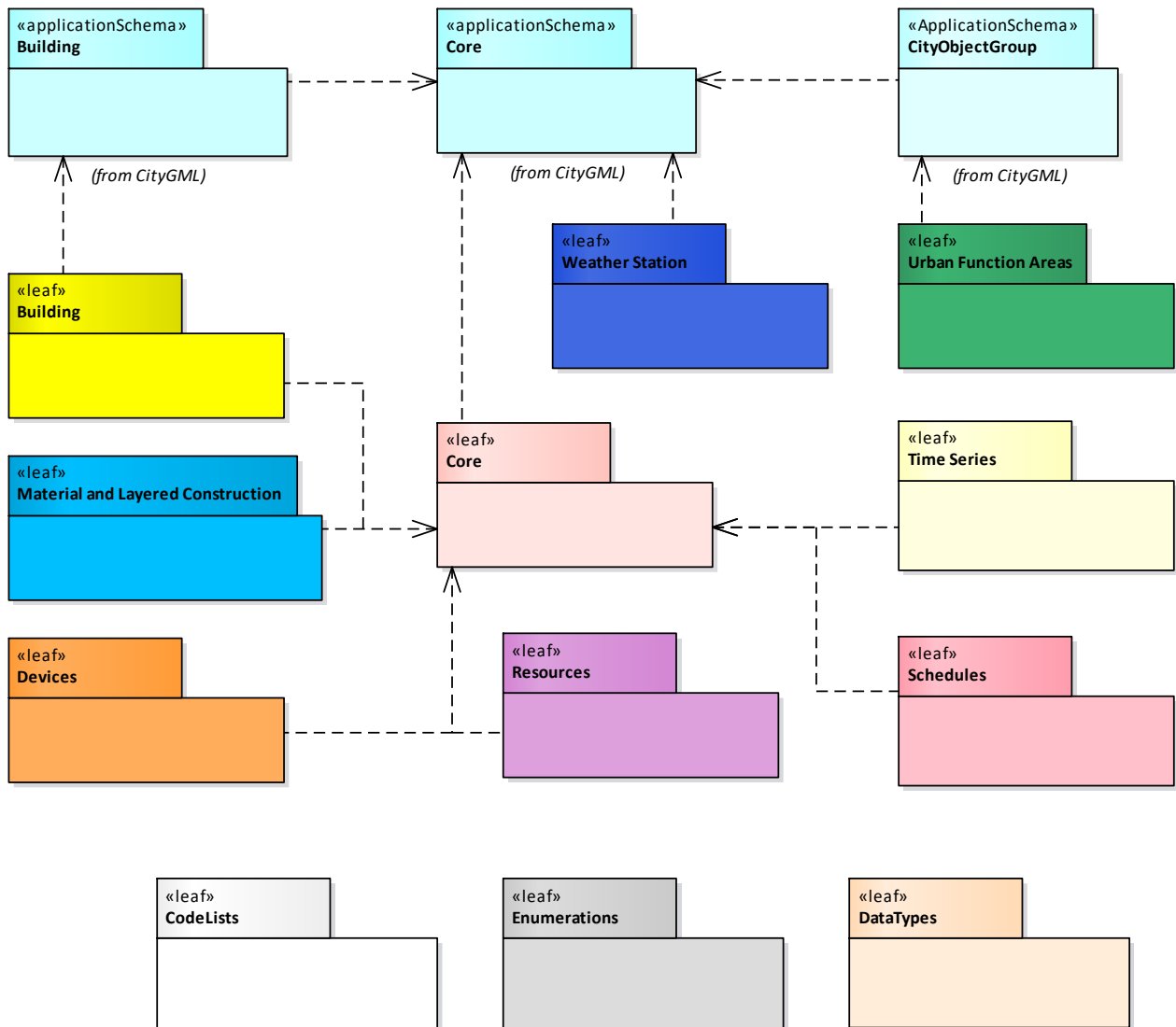
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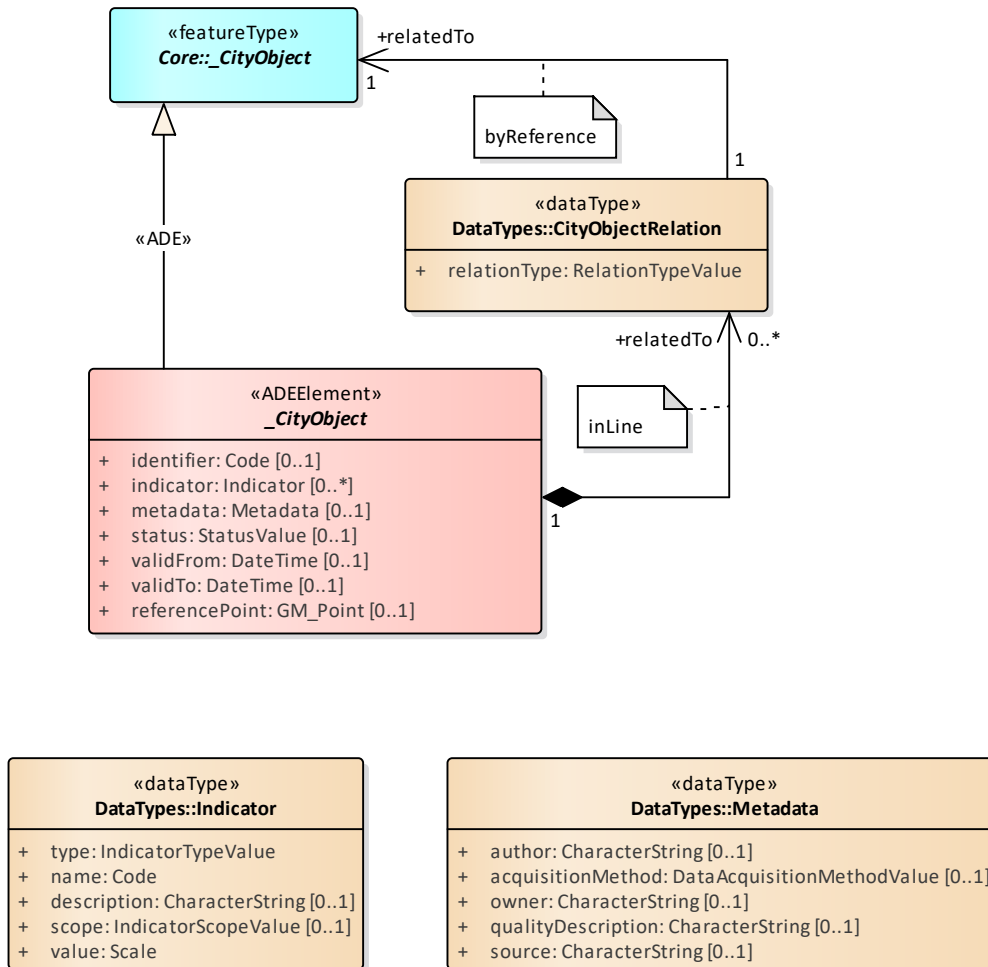
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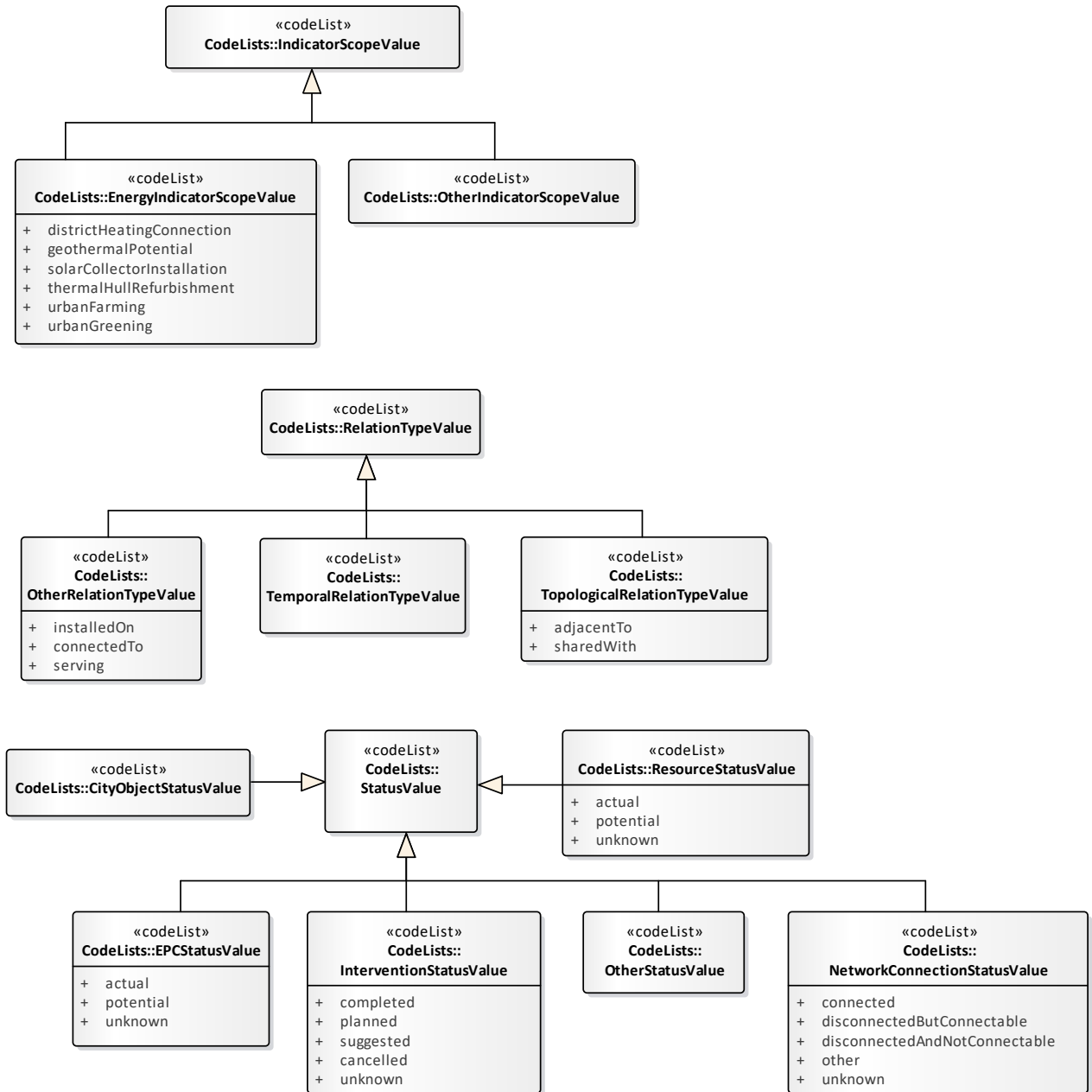
Overview of the packages



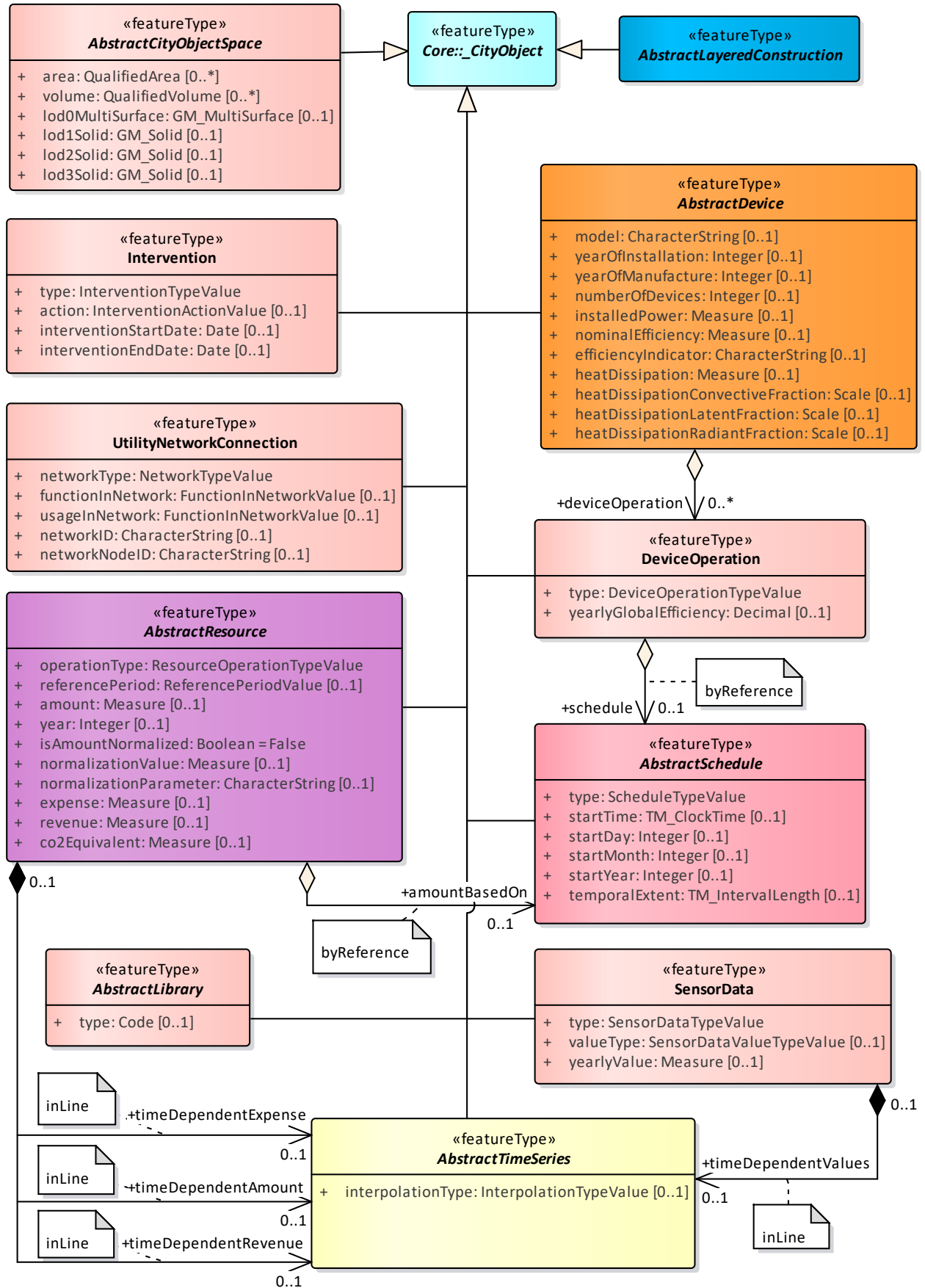
Core package part 1



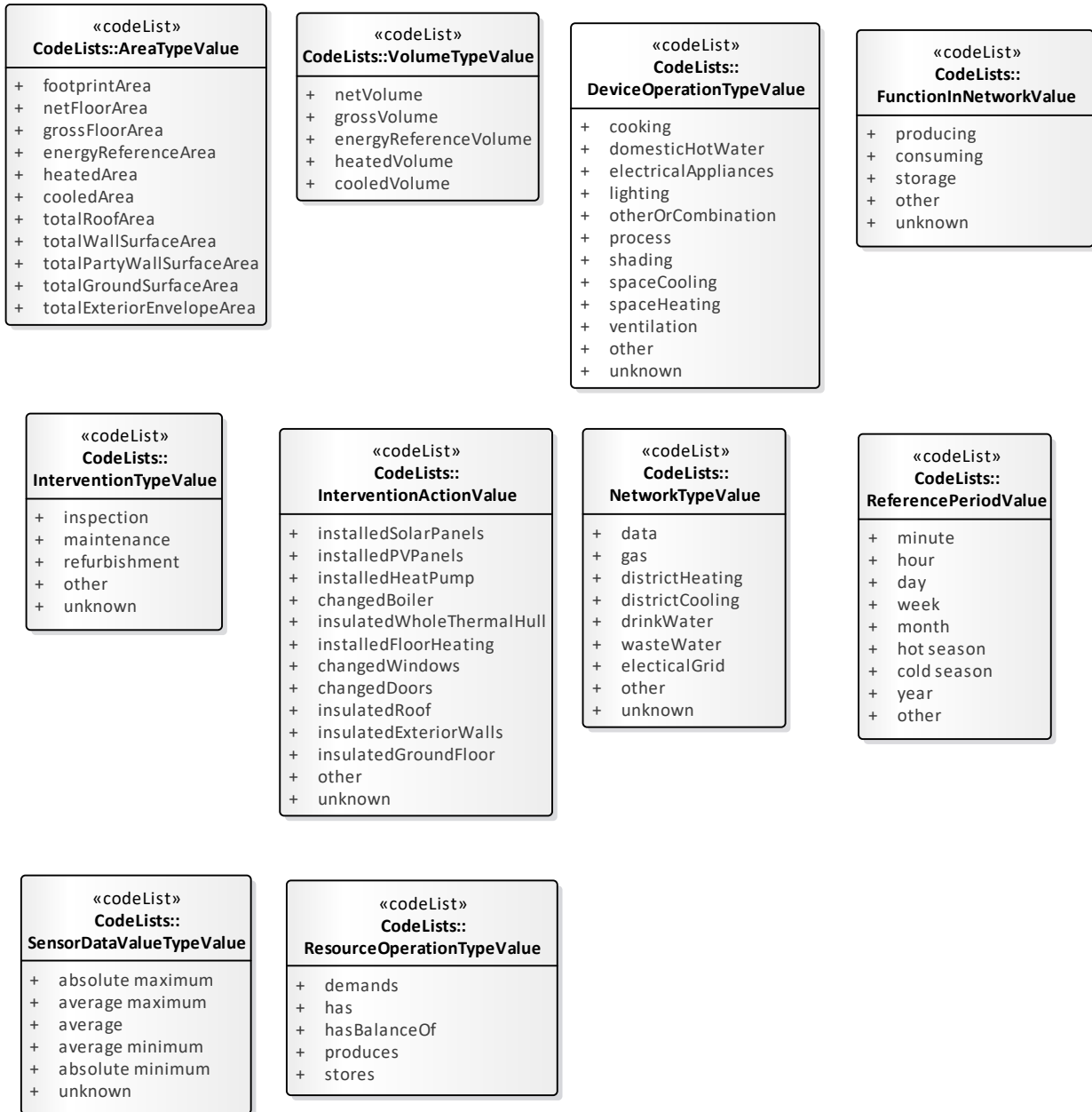
Core package part 1: CodeLists



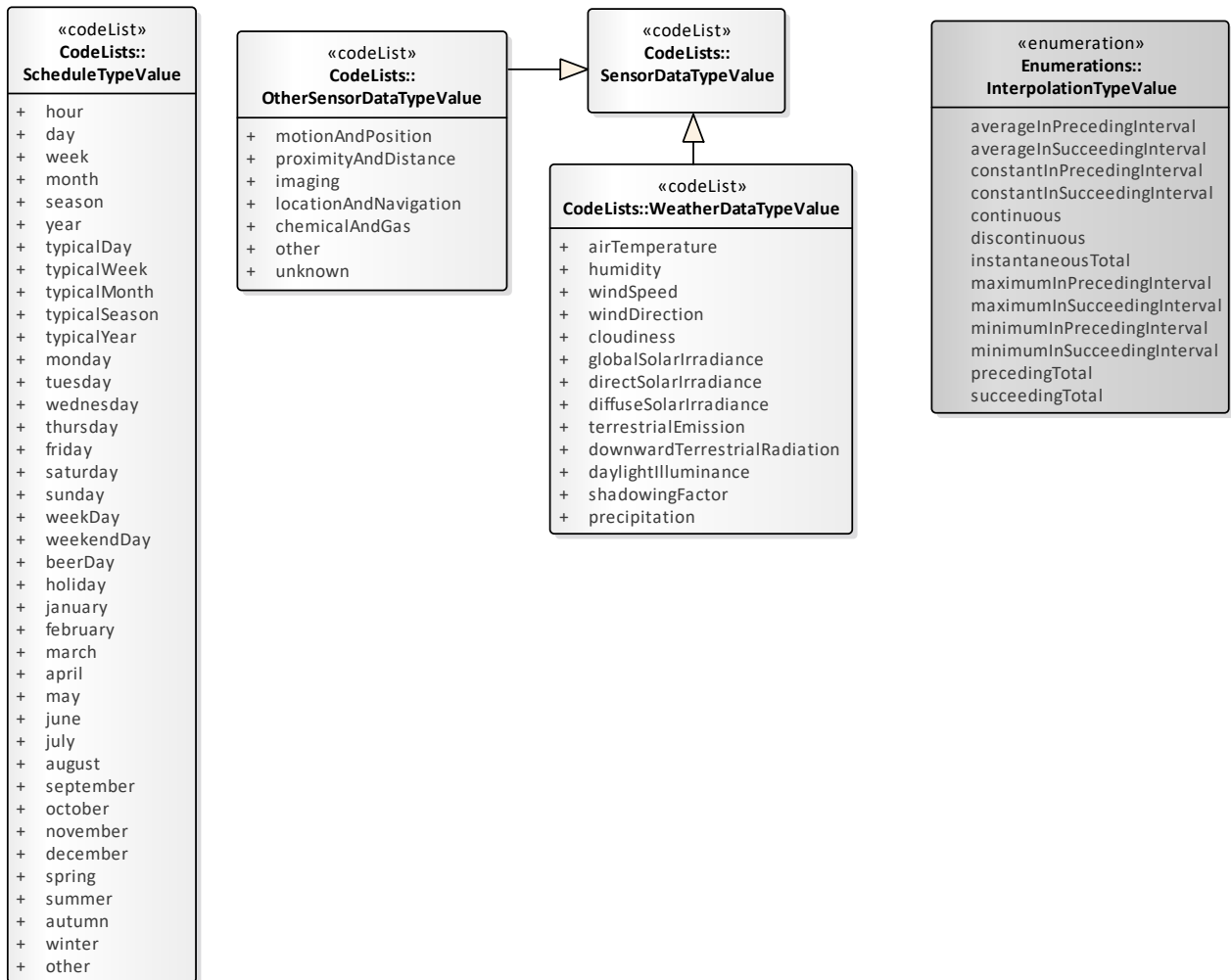
Core package part 2



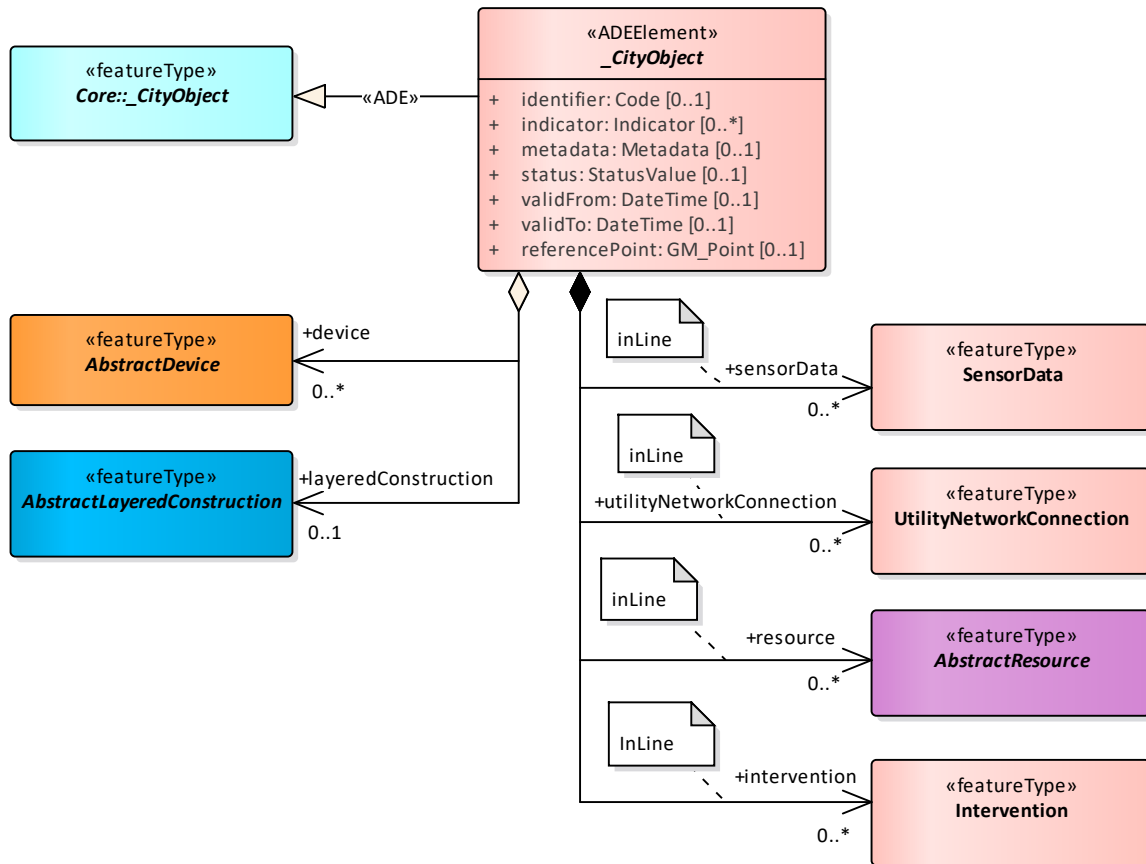
Core package part 2: CodeLists



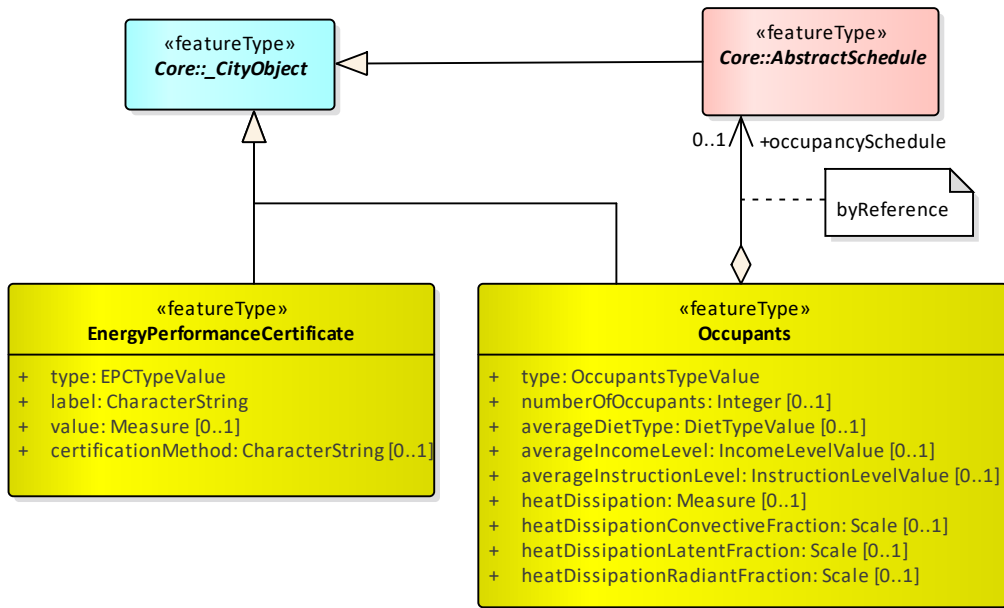
Core package part 2: CodeLists and Enumerations

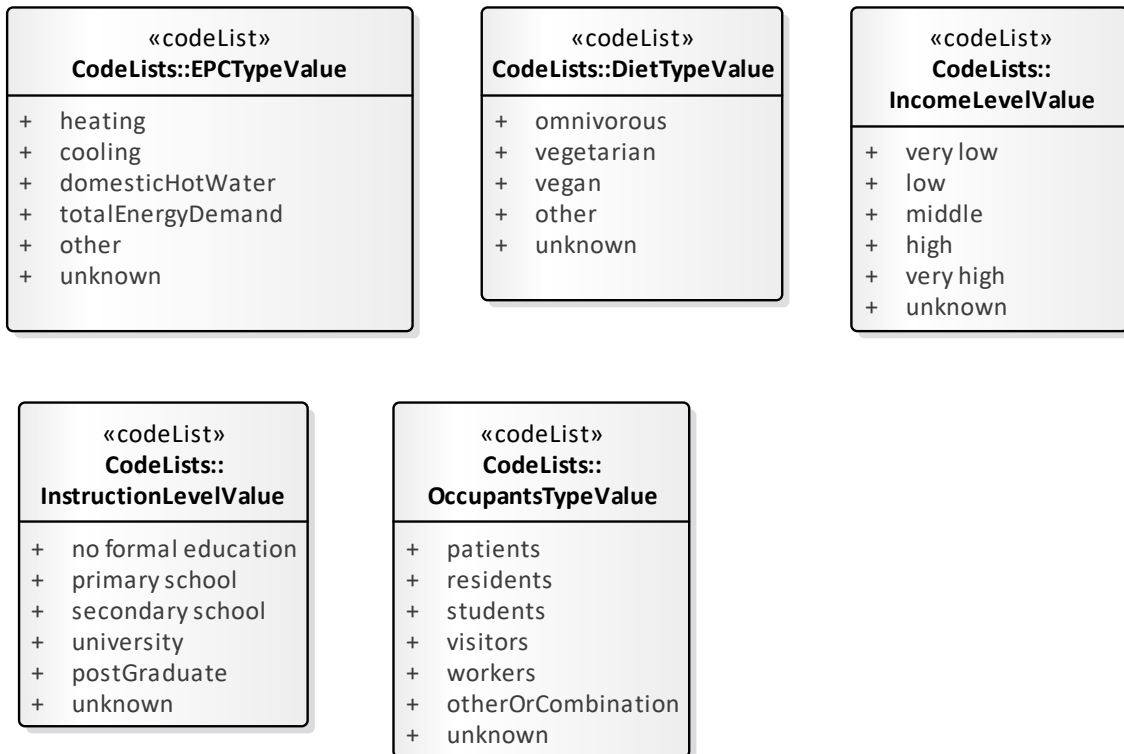


Core package part 3

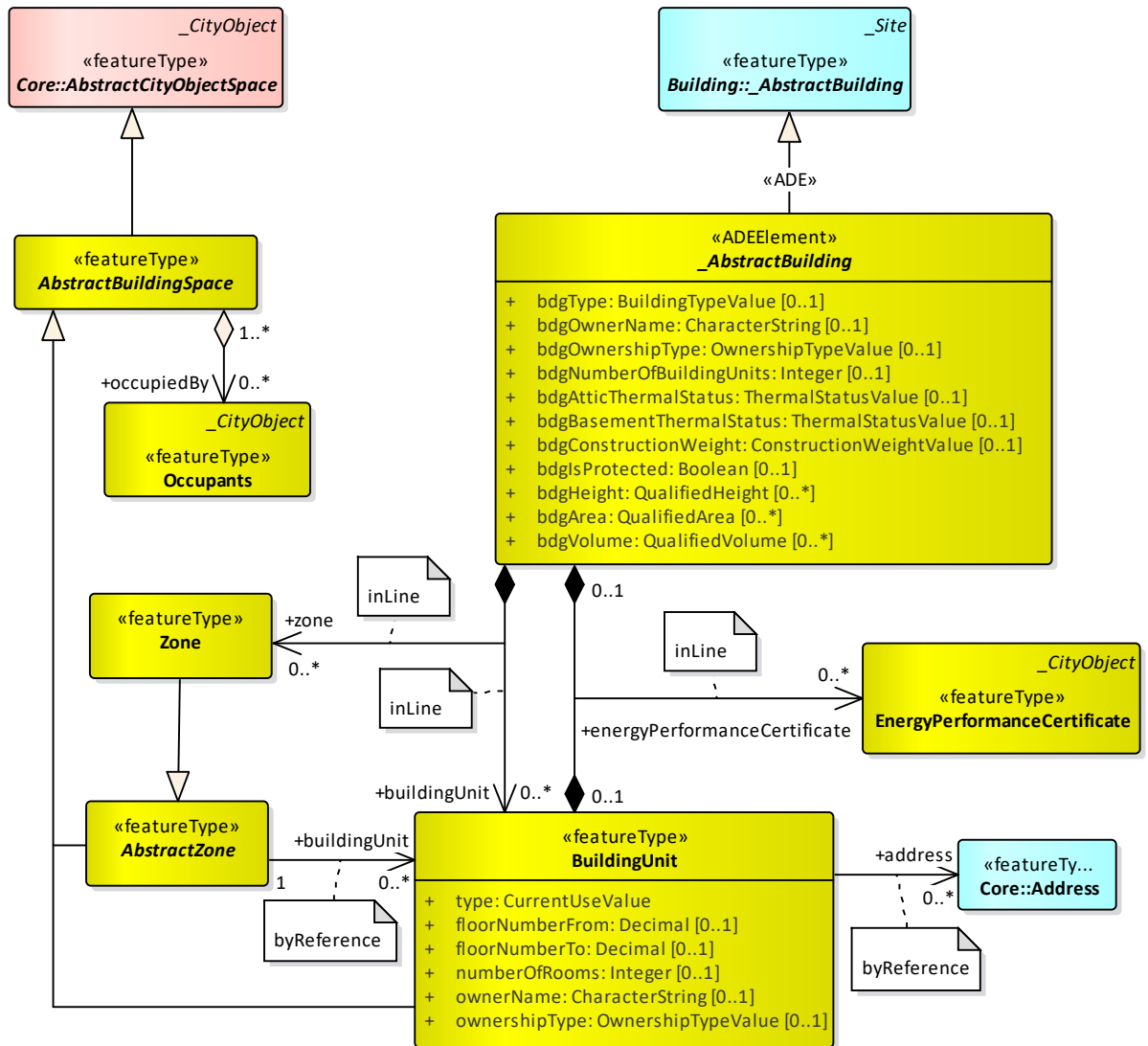


Building package part 1

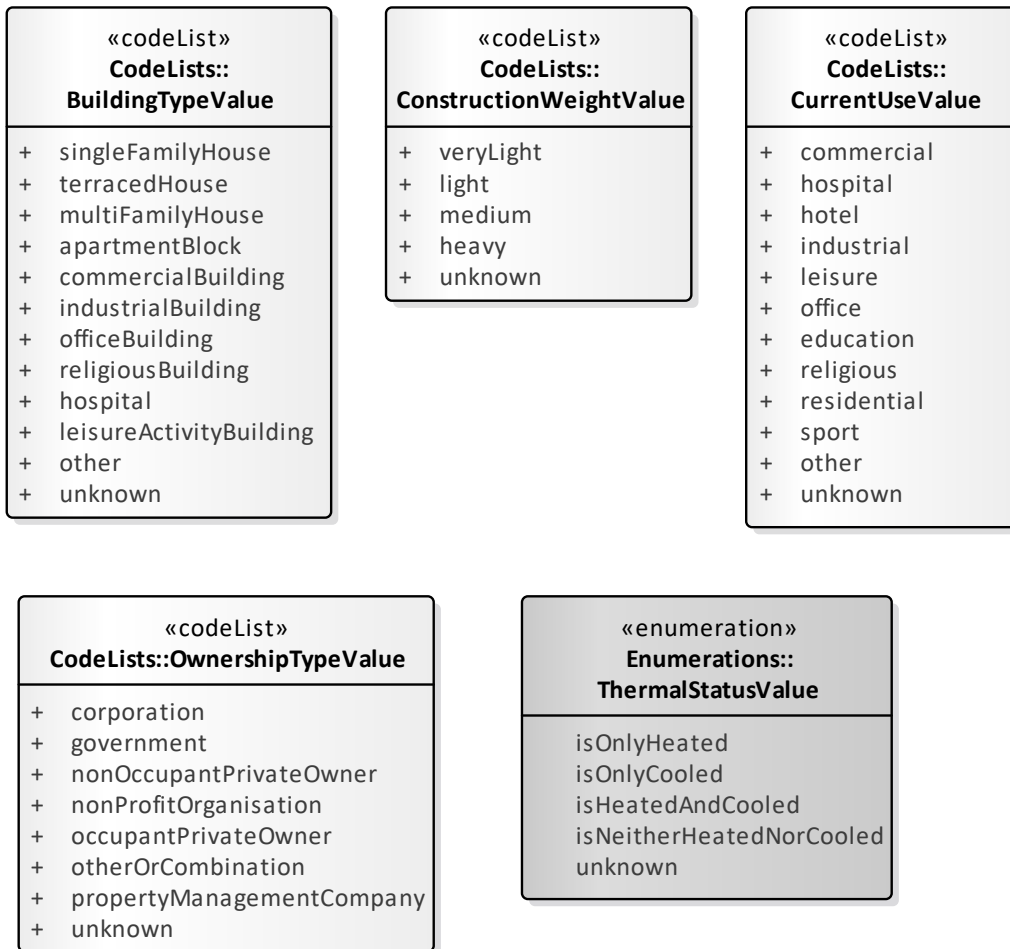


Building package part 1: CodeLists and Enumerations

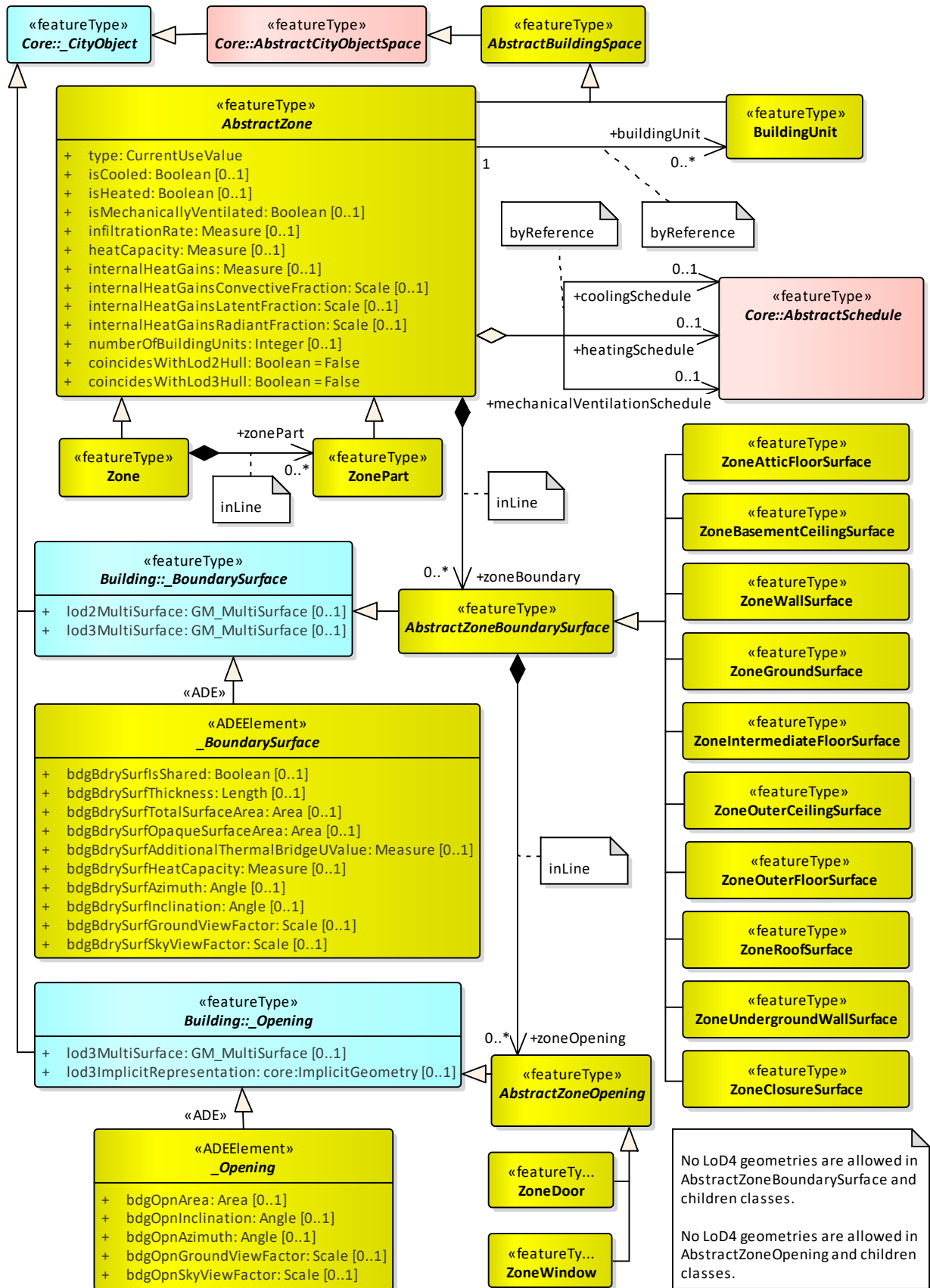
Building package part 2



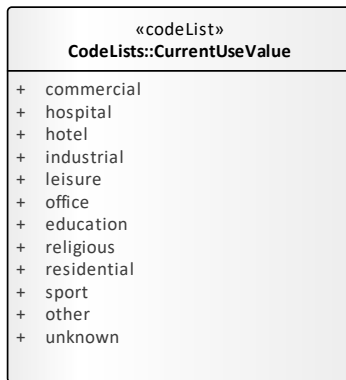
Building package part 2: CodeLists and Enumerations



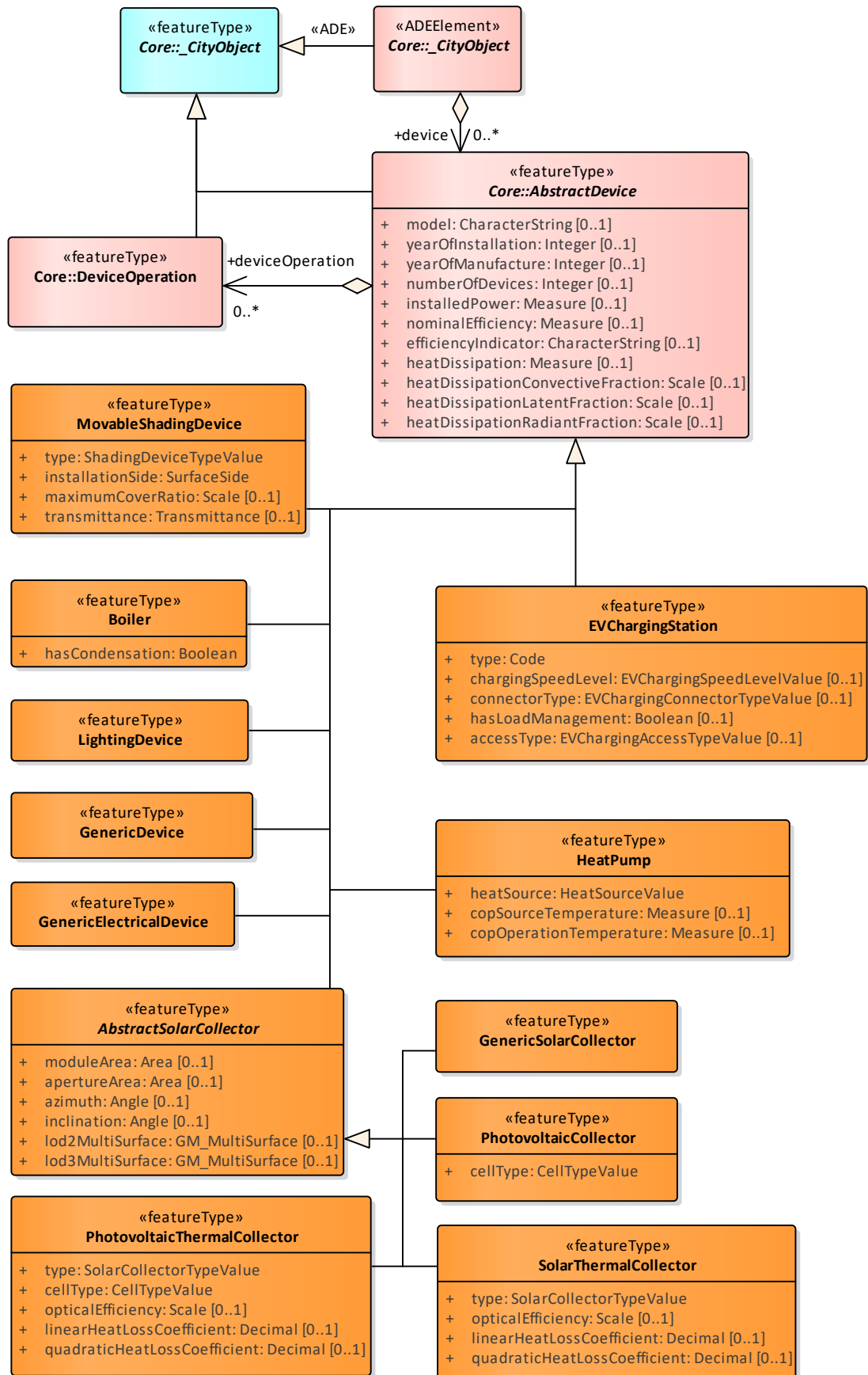
Building package part 3



Building package part 3: CodeLists and Enumerations



Devices package part 1



Devices package part 1: CodeLists and Enumerations

«codeList» CodeLists::CellTypeValue
+ monocrystalline + polycrystalline + amorphous + other + unknown

«codeList» CodeLists::EVChargingAccessValue
+ private + public + unknown

«codeList» CodeLists::EVChargingConnectorTypeValue
+ AC - J1772 (Type1) + AC - Mennekes (Type2) + AC - GB/T + AC/DC - Tesla + DC - CHAdeMO + DC - CCS1 + DC - CCS2 + DC - GB/T + unknown

«codeList» CodeLists::EVChargingSpeedLevelValue
+ Level 1 + Level 2 + Level 3

«codeList» CodeLists::EVChargingStationTypeValue
+ AC + DC + unknown

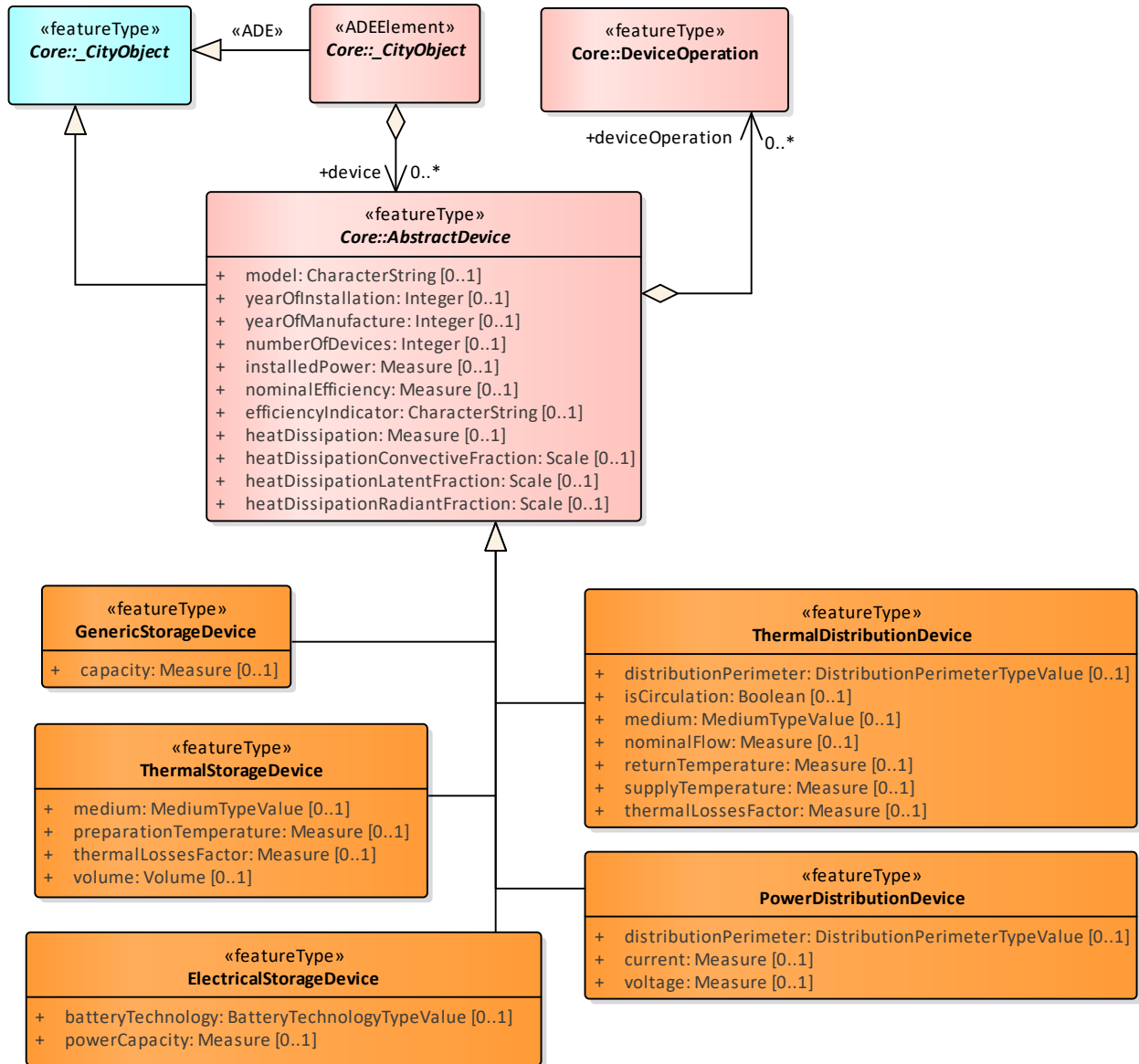
«codeList» CodeLists::HeatSourceValue
+ ambientAir + aquifer + exhaustAir + horizontalGroundCollector + verticalGroundCollector + unknown

«codeList» CodeLists::ShadingDeviceTypeValue
+ curtain + shutter + blind + other + unknown

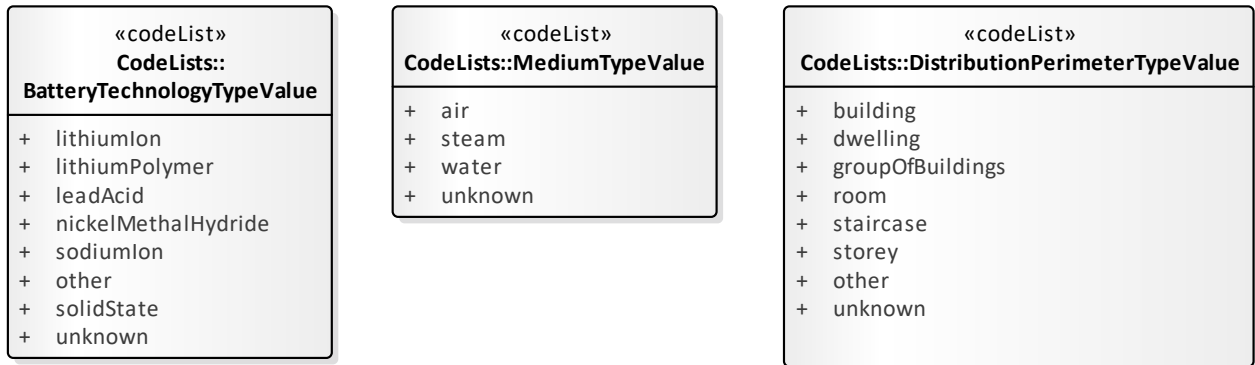
«codeList» CodeLists::SolarCollectorTypeValue
+ flatPlaneCollector + evacuatedTubeCollector + unknown

«enumeration» Enumerations::SurfaceSide
inside outside

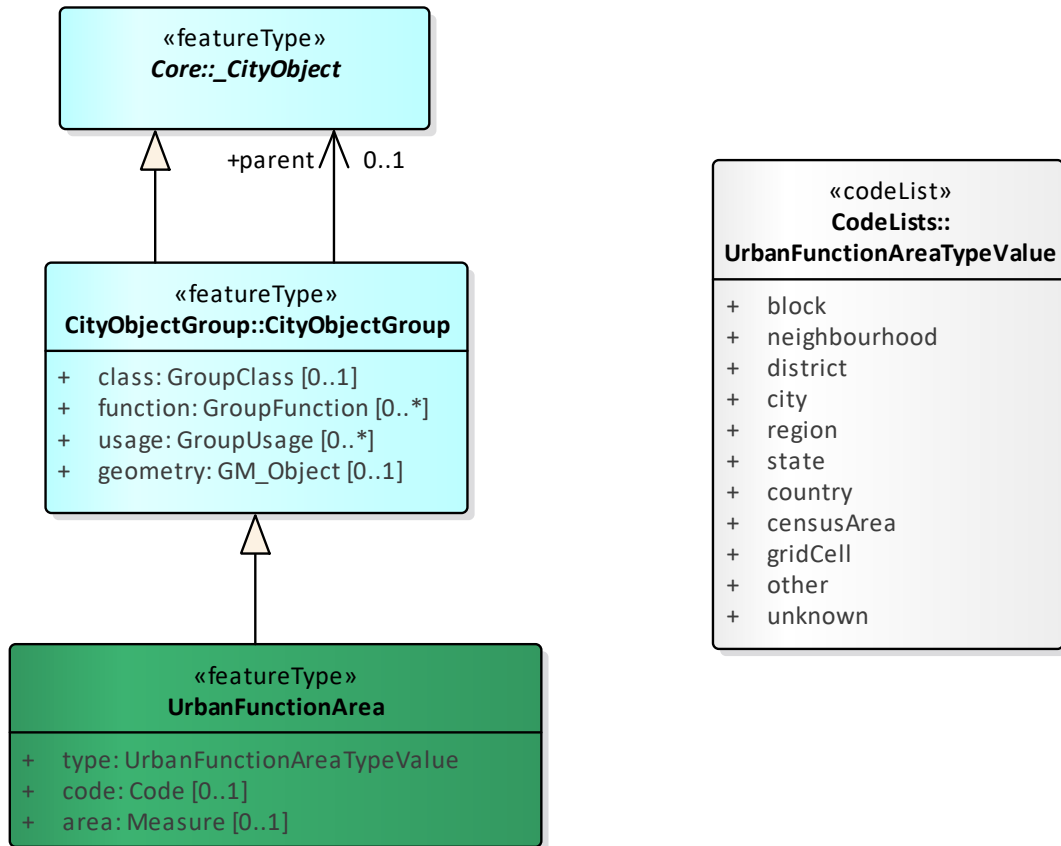
Devices package part 2



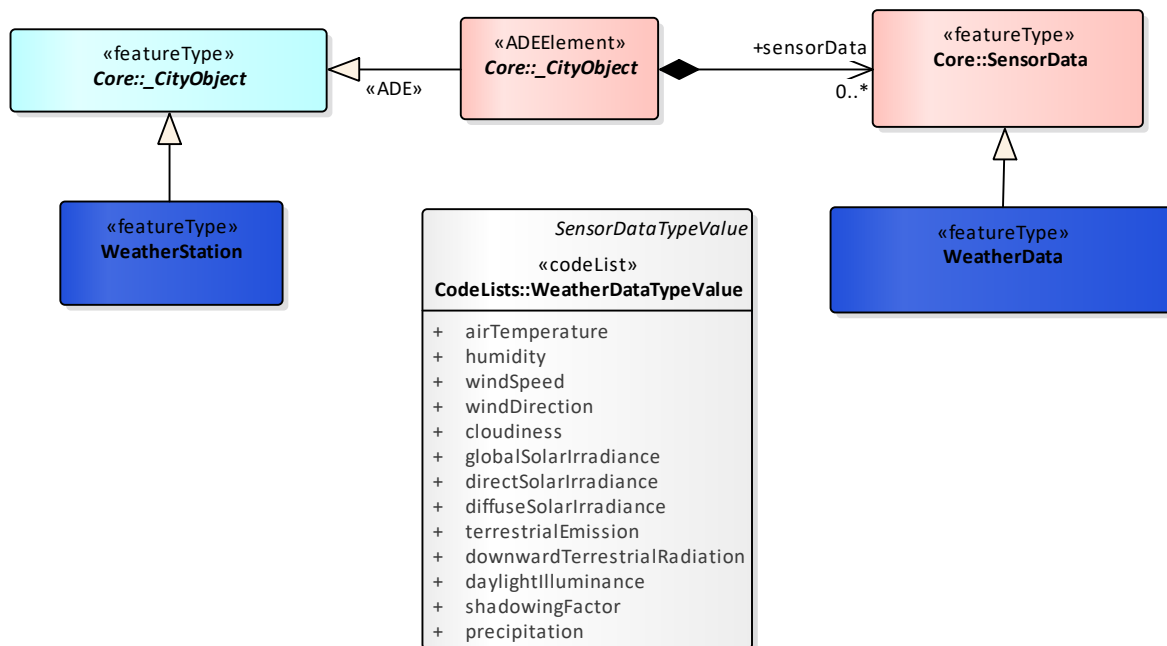
Devices package part 2: CodeLists and Enumerations



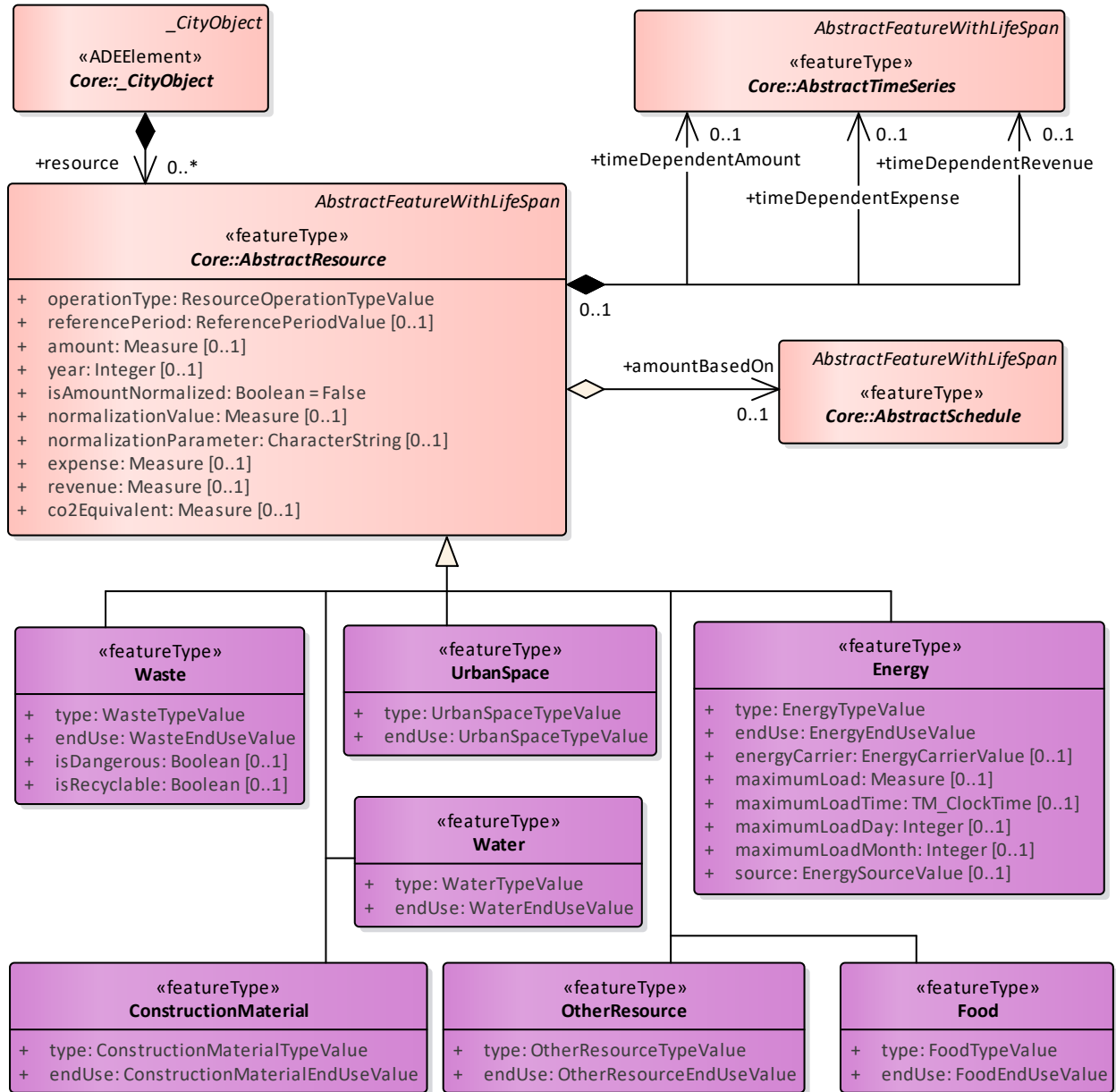
Urban function areas package and CodeLists



Weather station package and CodeLists



Resources package



Resources package: CodeLists and Enumerations

«codeList» CodeLists:: ConstructionMaterialEndUseValue
+ construction
+ refurbishment
+ otherOrCombination
+ unknown

«codeList» CodeLists:: ConstructionMaterialTypeValue
+ wood
+ tiles
+ concrete
+ sand
+ soil
+ otherOrCombination
+ glass
+ bricks
+ unknown

«codeList» CodeLists:: EnergyCarrierValue
+ electricity
+ coal
+ naturalGas
+ chilledAir
+ chilledWater
+ fuelOil
+ hotAir
+ steam
+ hotWater
+ woodChips
+ hydrogen
+ propane
+ woodPallets
+ other
+ unknown

«codeList» CodeLists:: EnergyEndUseValue
+ cooking
+ domesticHotWater
+ electricalAppliances
+ lighting
+ mobility
+ otherOrCombination
+ spaceCooling
+ spaceHeating
+ ventilation
+ process
+ construction
+ unknown

«codeList» CodeLists:: EnergyTypeValue
+ actual
+ net
+ primary
+ secondary
+ final
+ useful

«codeList» CodeLists:: EnergySourceValue
+ photovoltaicPanels
+ powerGrid
+ districtHeating
+ solarThermalPanels
+ heatPump
+ wind

«codeList» CodeLists:: FoodEndUseValue
+ alimentation
+ commerce

«codeList» CodeLists:: FoodTypeValue
+ greeneries
+ meat
+ cereals
+ cheese
+ milk
+ wine
+ beer
+ unknown

«codeList» CodeLists:: OtherResourceEndUseValue

«codeList» CodeLists:: OtherResourceTypeValue

«codeList» CodeLists:: UrbanSpaceEndUseValue
--

«codeList» CodeLists:: UrbanSpaceTypeValue
--

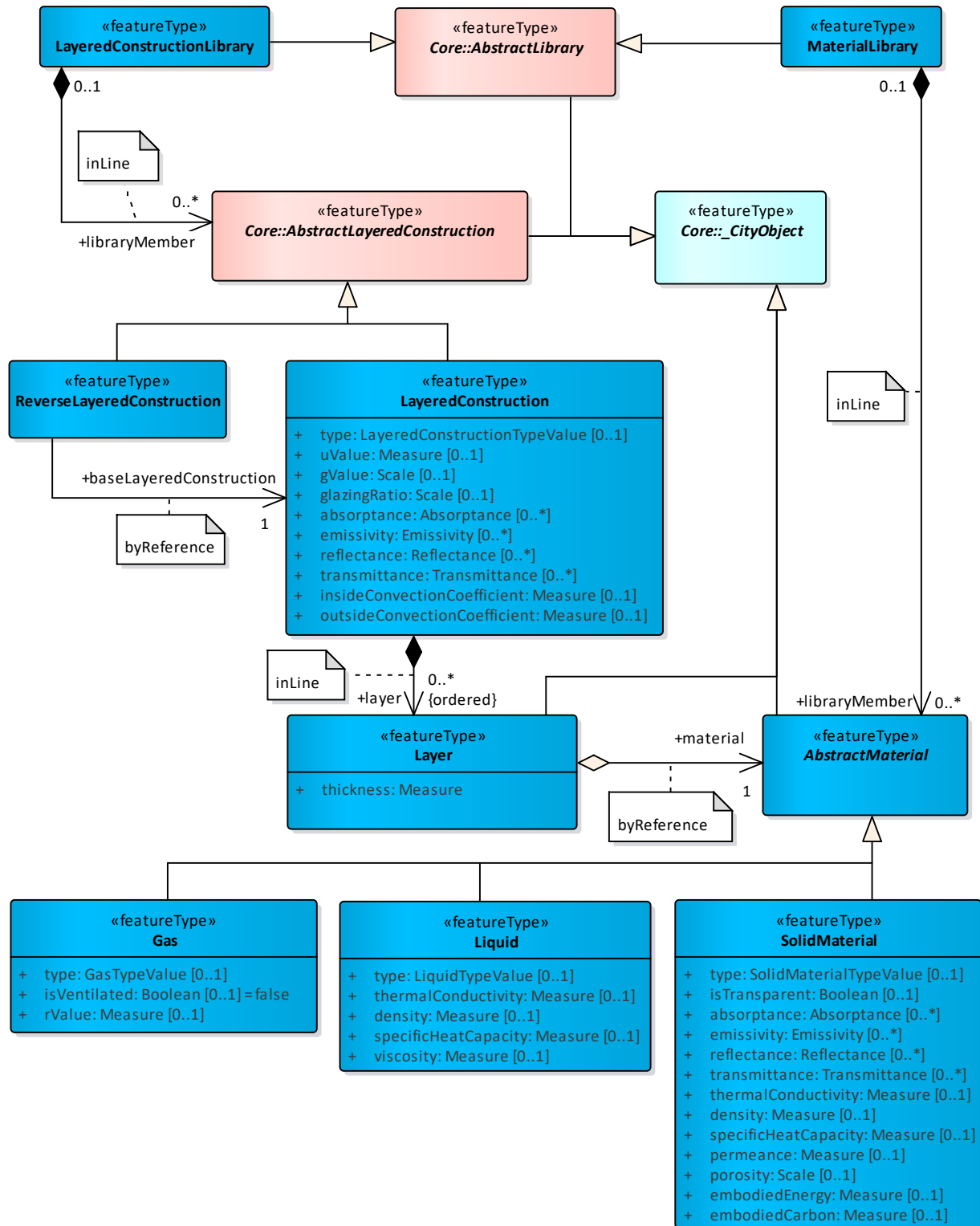
«codeList» CodeLists:: WasteEndUseValue
+ incineration
+ landFilling
+ recycling
+ other
+ unknown

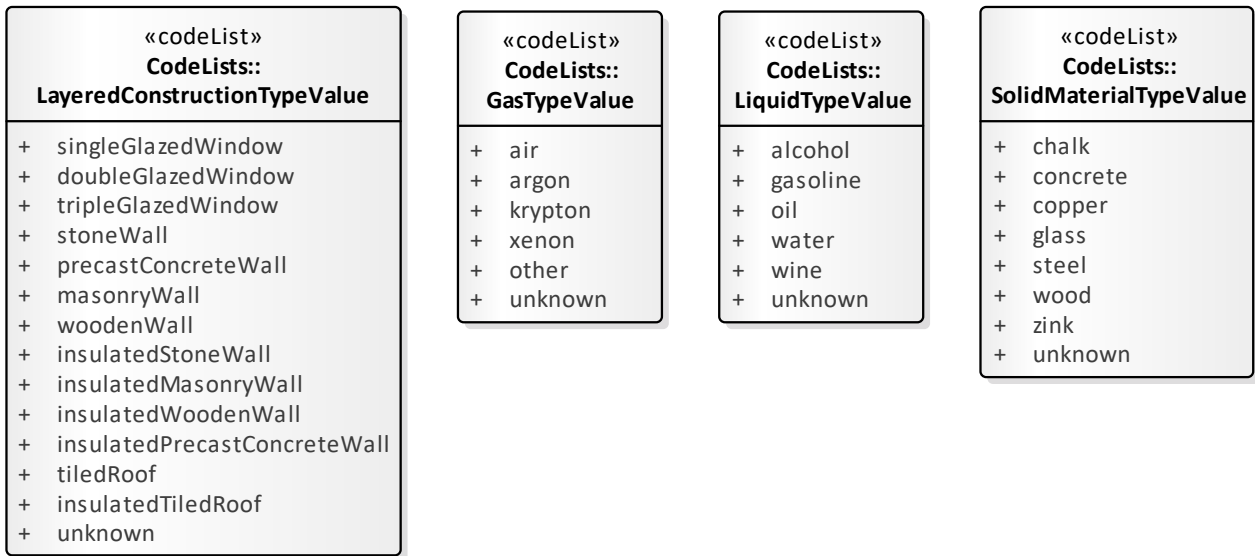
«codeList» CodeLists:: WasteTypeValue
+ organic
+ plastic
+ glass
+ paper
+ metal
+ water
+ otherOrCombination
+ unknown

«codeList» CodeLists:: WaterEndUseValue
+ drinking
+ washing
+ processing
+ otherOrCombination
+ unknown

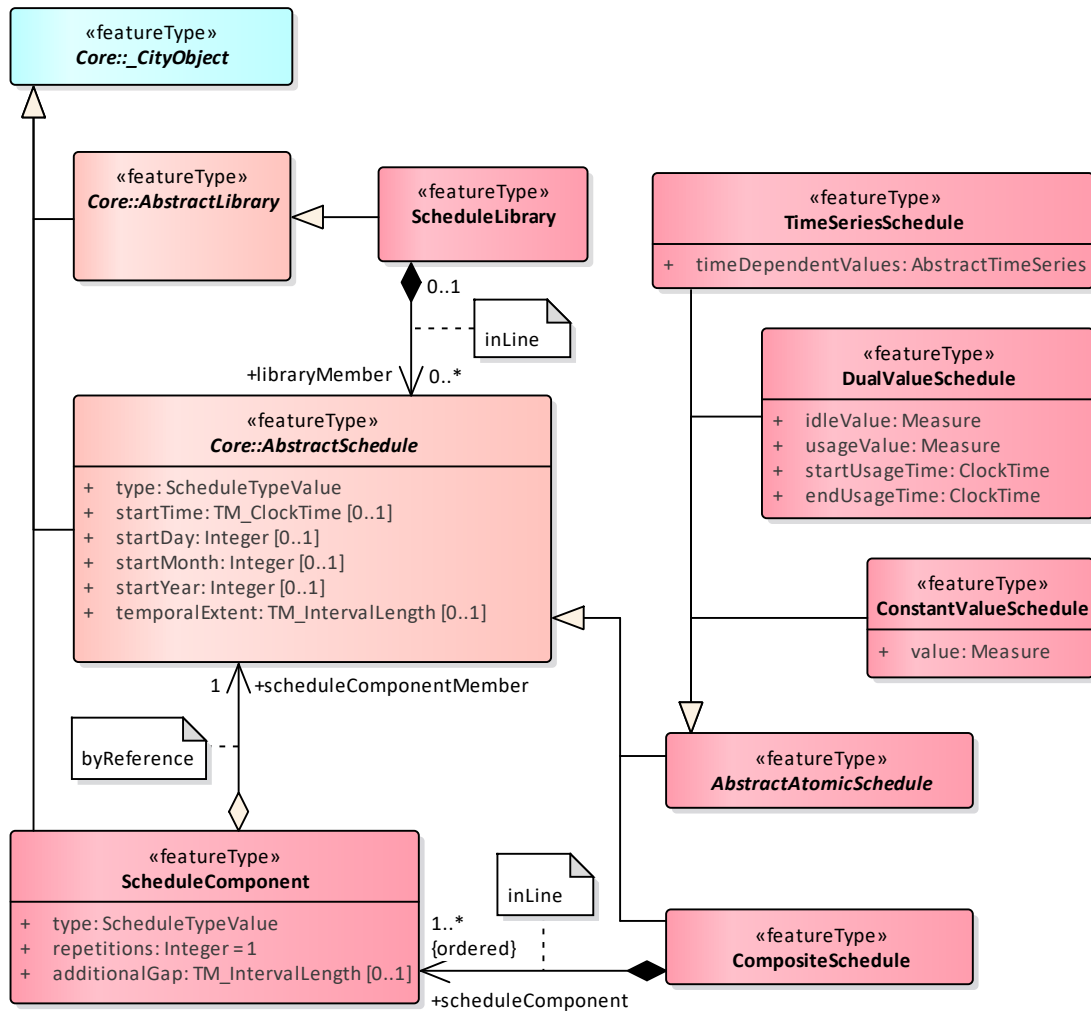
«codeList» CodeLists:: WaterTypeValue
+ drinkWater
+ wasteWater
+ rainWater
+ otherOrCombination
+ unknown

Layered construction package



Layered construction package: CodeLists

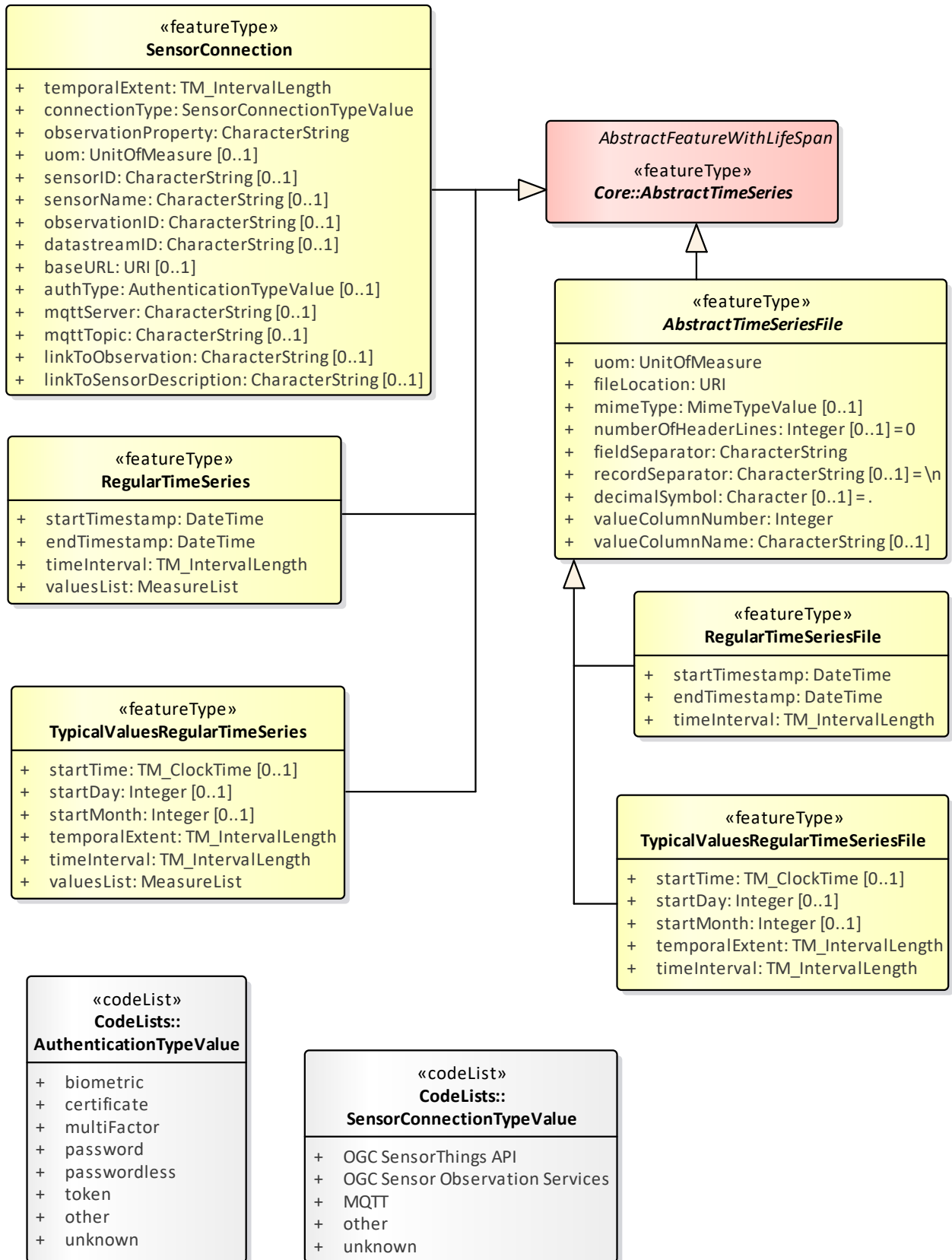
Schedules package



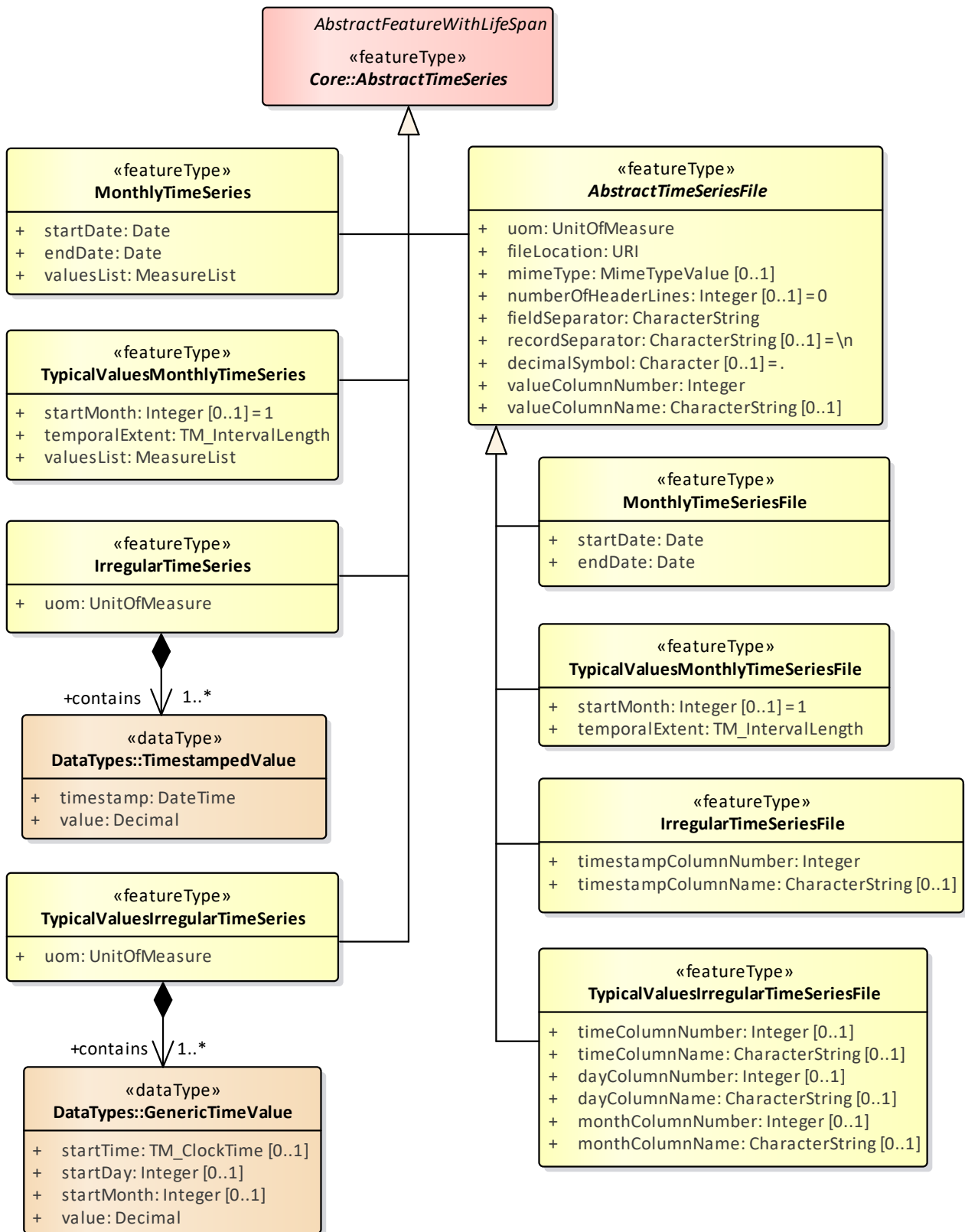
Schedules package: CodeLists

«codeList» CodeLists:: ScheduleTypeValue
+ hour
+ day
+ week
+ month
+ season
+ year
+ typicalDay
+ typicalWeek
+ typicalMonth
+ typicalSeason
+ typicalYear
+ monday
+ tuesday
+ wednesday
+ thursday
+ friday
+ saturday
+ sunday
+ weekDay
+ weekendDay
+ beerDay
+ holiday
+ january
+ february
+ march
+ april
+ may
+ june
+ july
+ august
+ september
+ october
+ november
+ december
+ spring
+ summer
+ autumn
+ winter
+ other

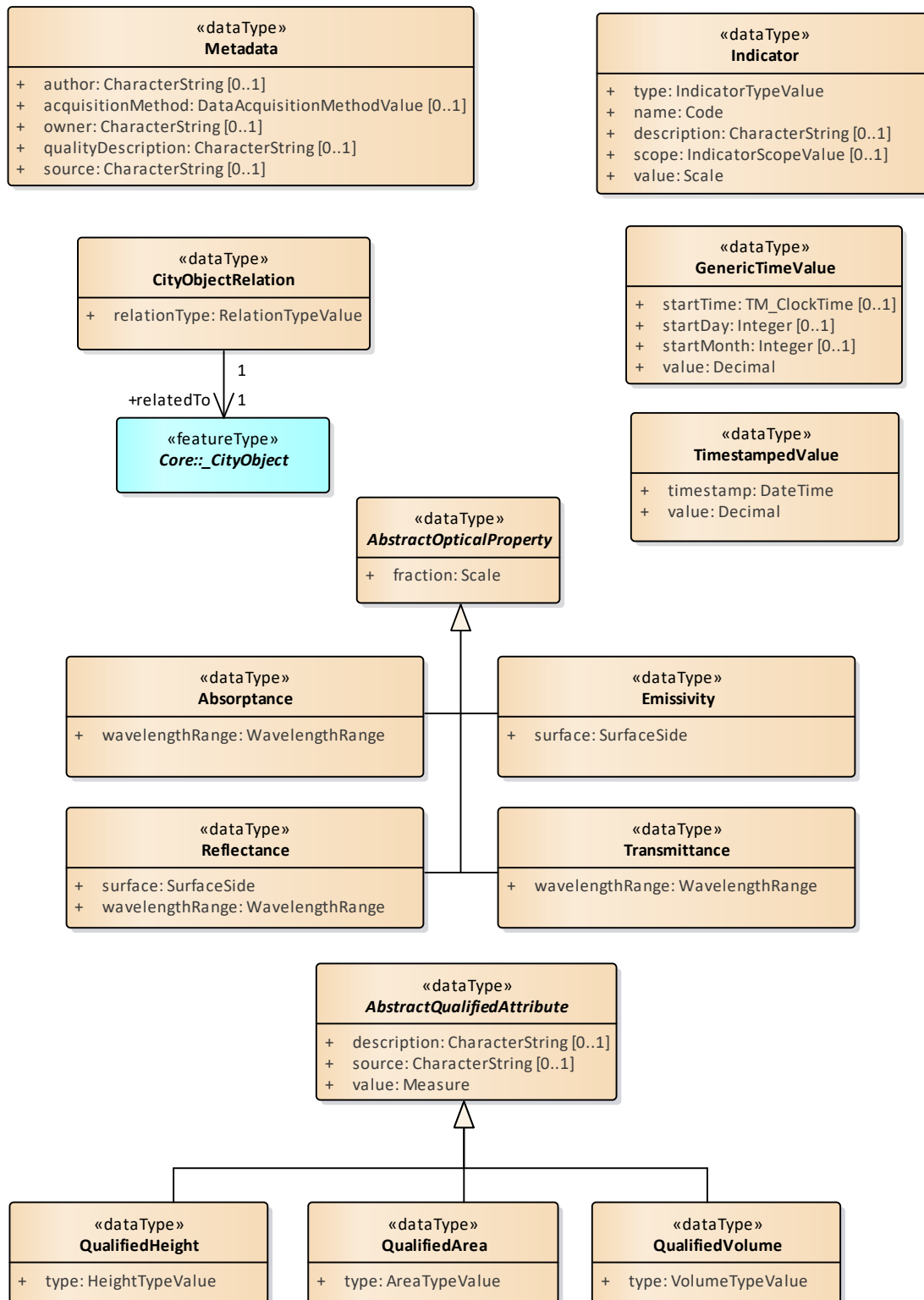
Time series package part 1 and CodeLists



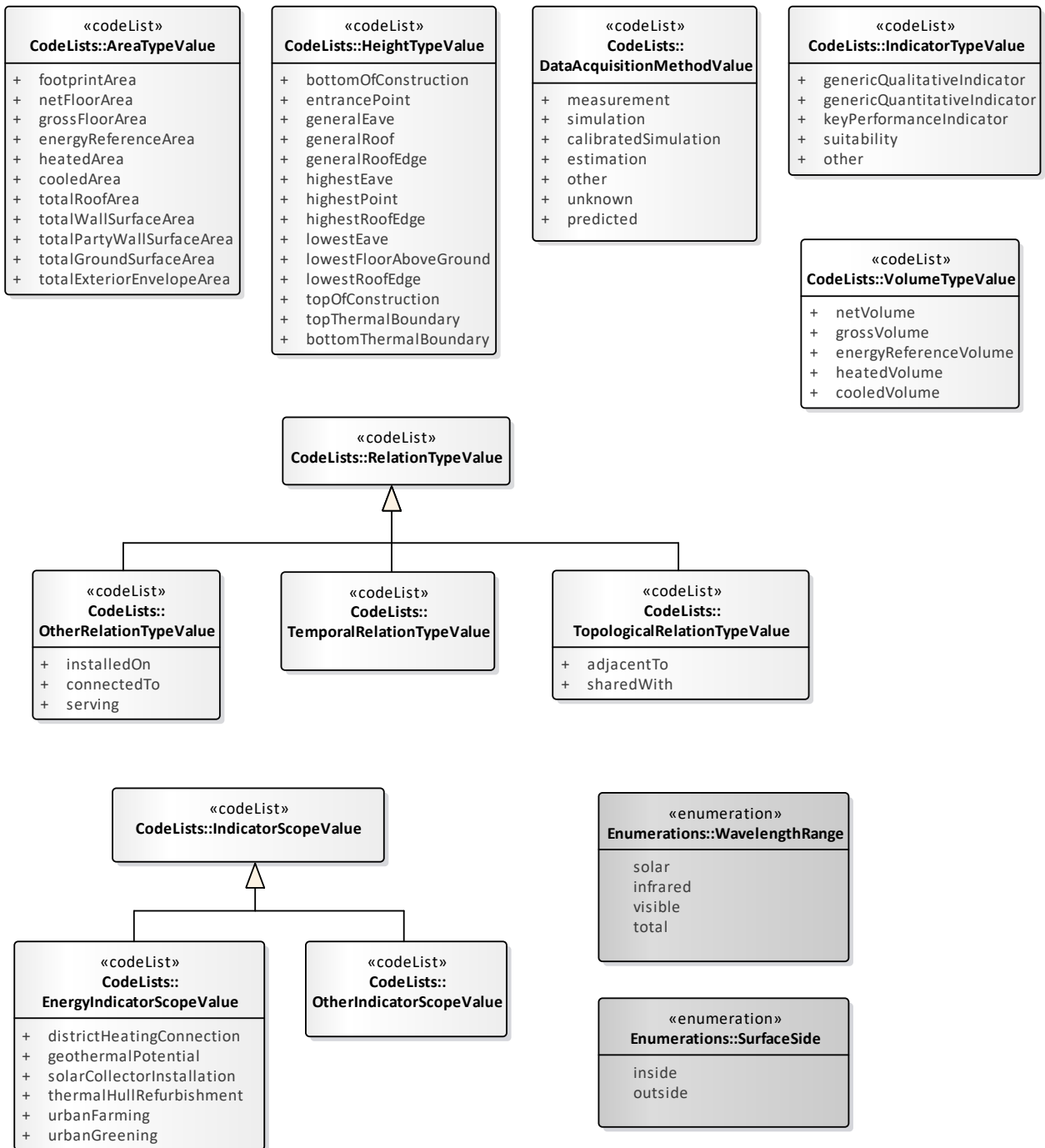
Time series package part 2



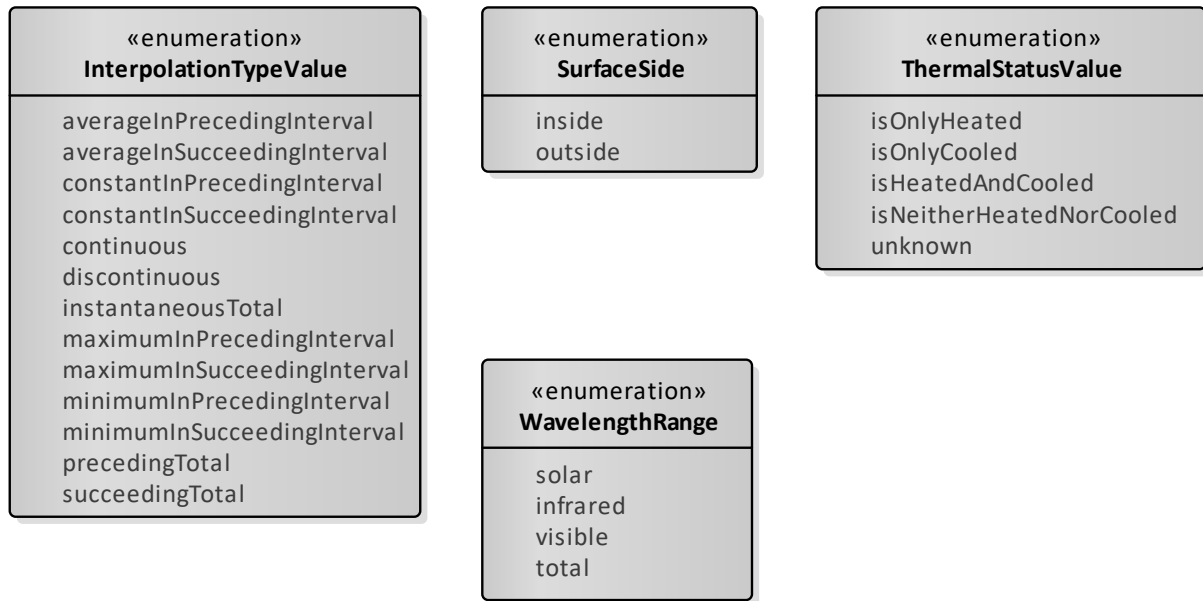
DataTypes package



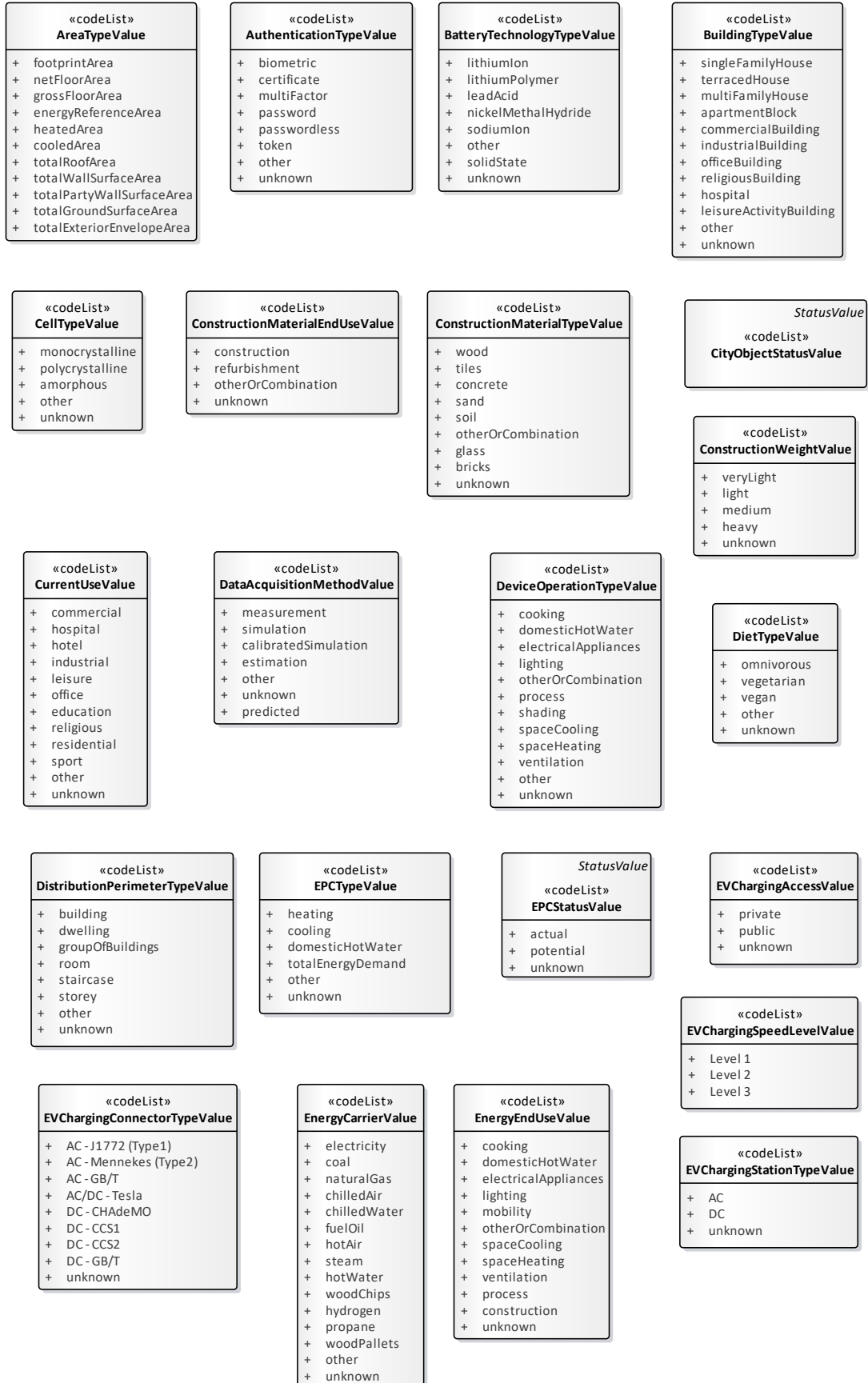
DataTypes package: CodeLists and Enumerations

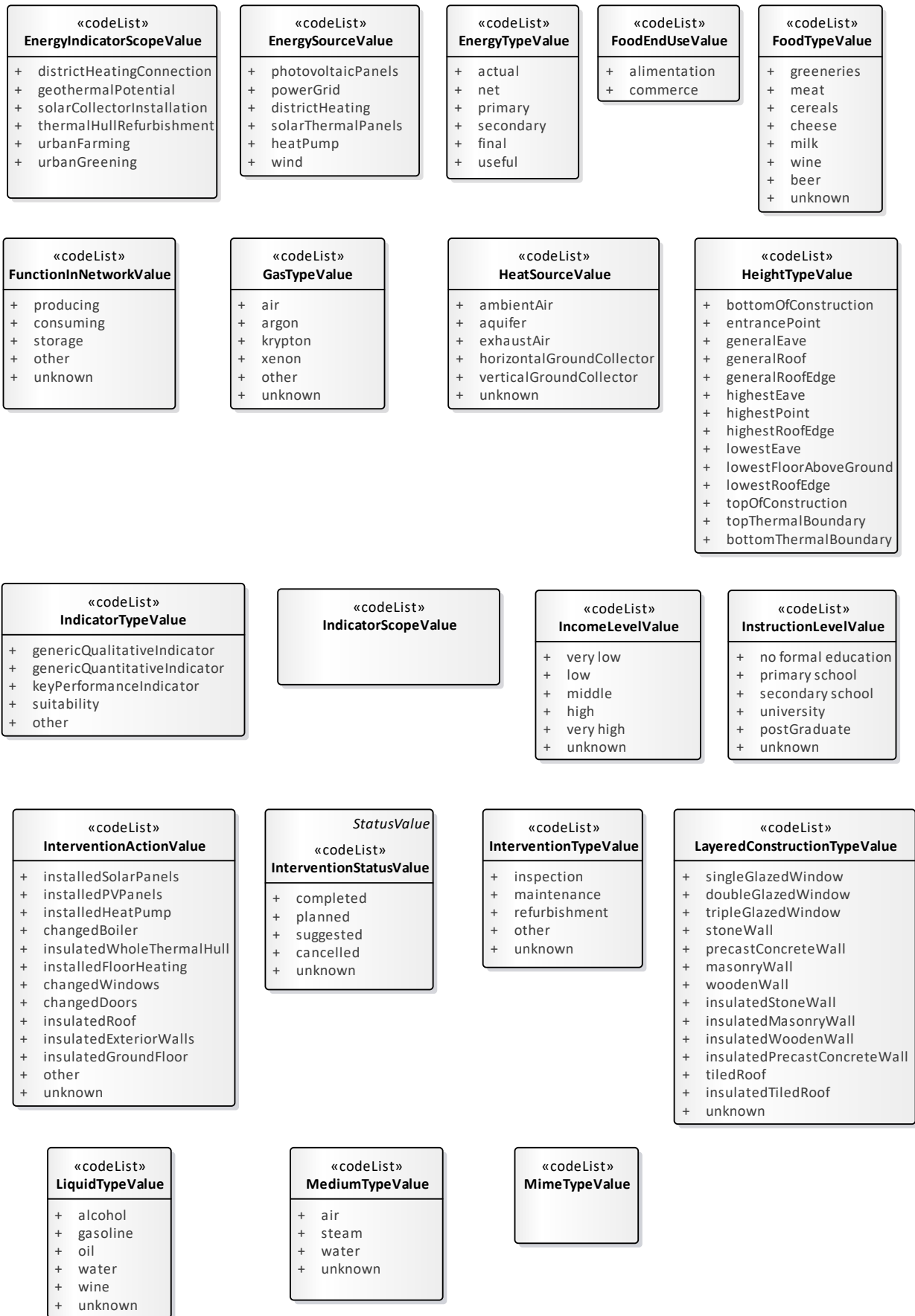


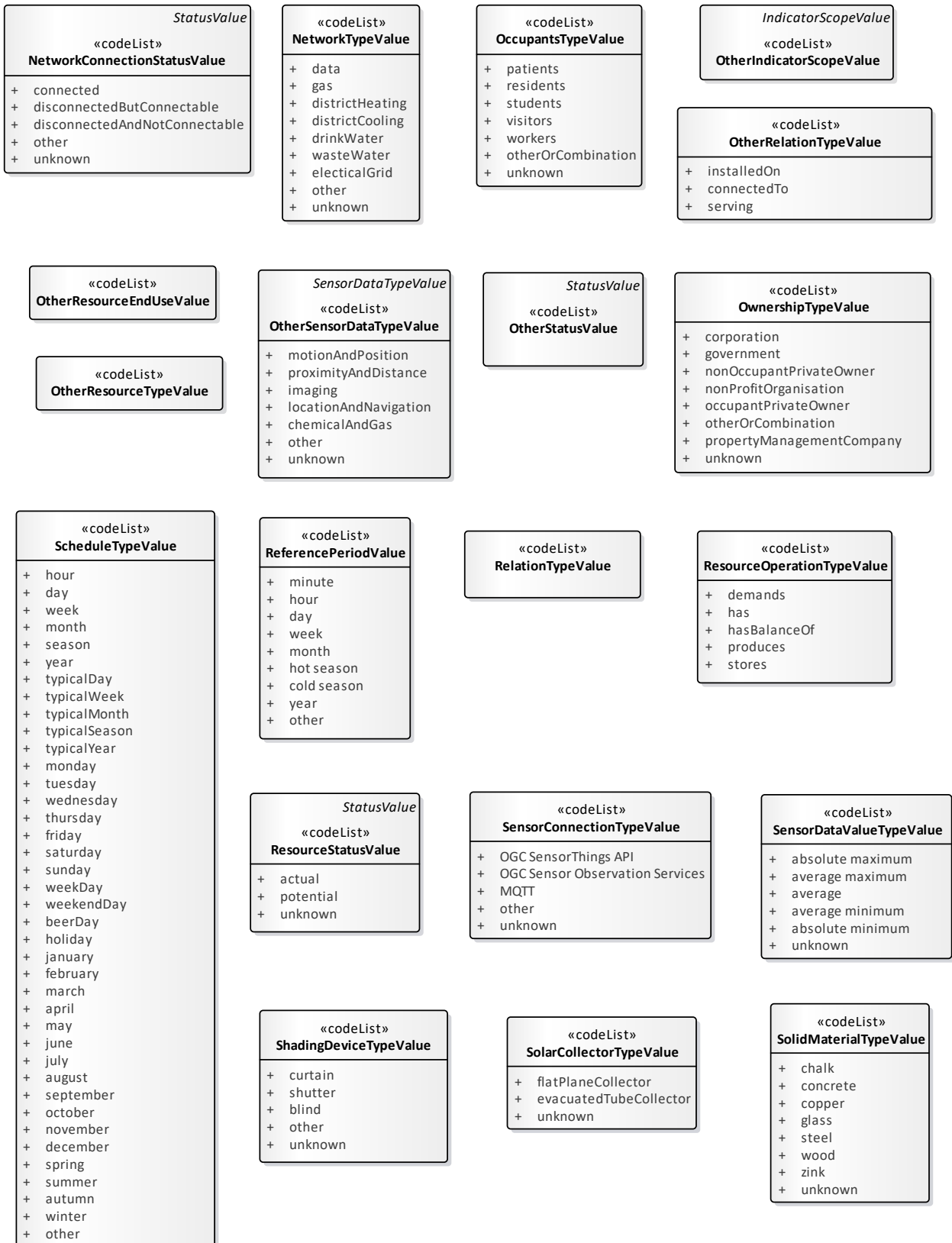
Enumerations package

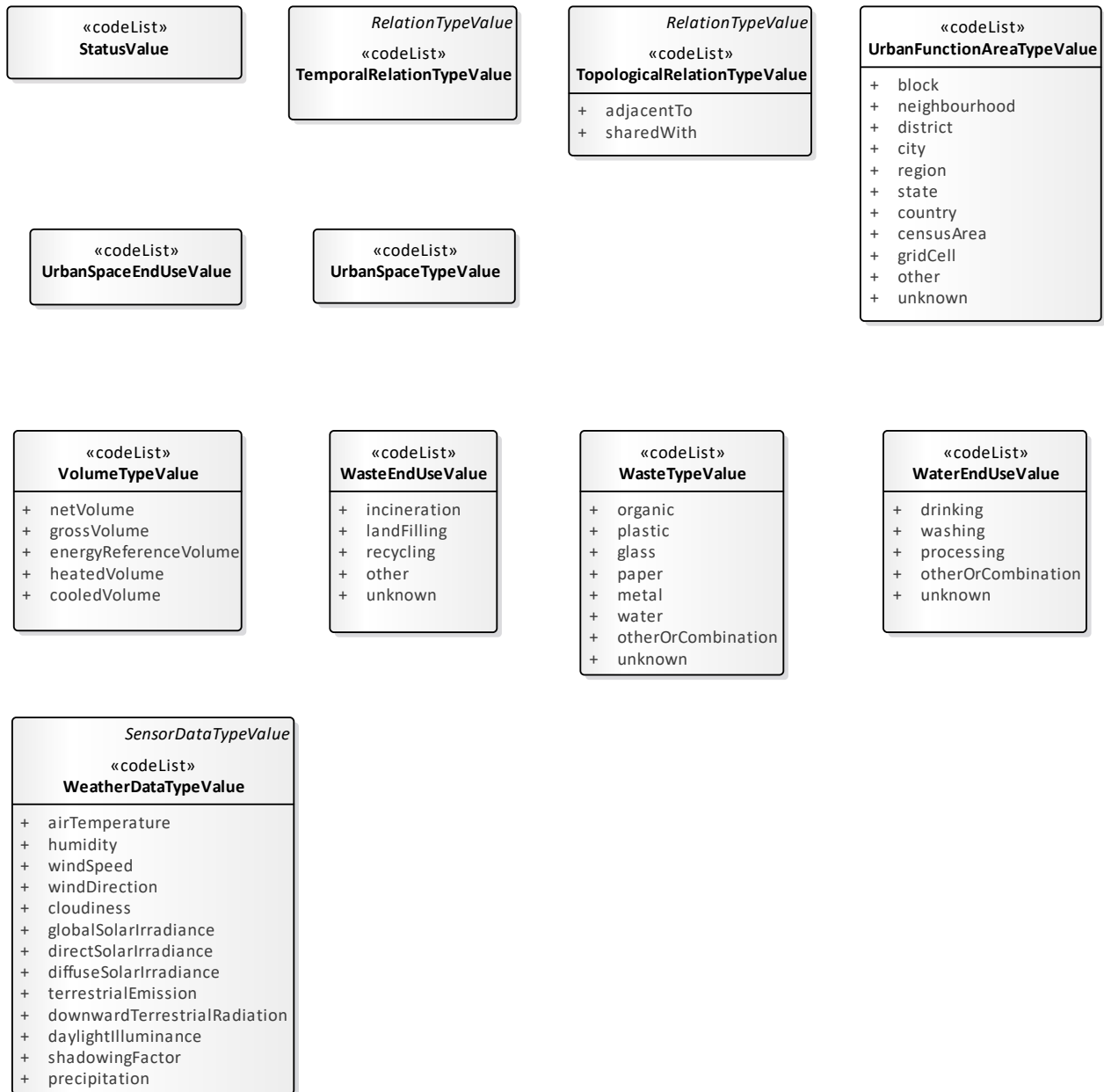


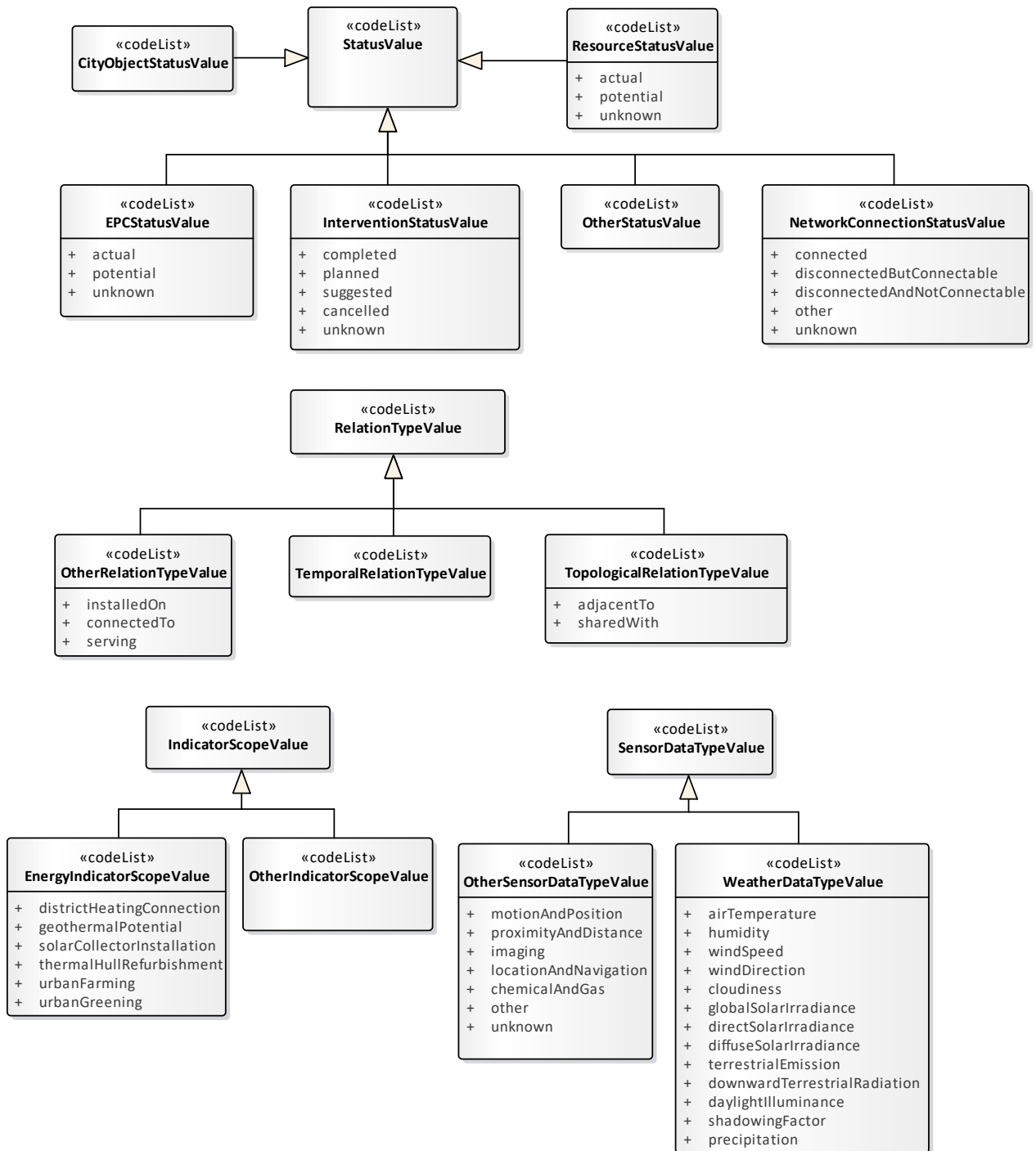
Codelists package











Changelog

Version 3.0 beta 8

2026-08-25

Building package:

- Class **ADE_AbstractBuilding**
 - - Changed target of property *zone* from **AbstractZone** to **Zone**

Layered Construction package

- Class **SolidMaterial**
 - Changed multiplicity of property *isTransparent* to 0..1

2026-08-08

Core package:

- Class **SensorData**:
 - Dropped property *position*

Devices package:

- Renamed class **PowerDistrubution** to **PowerDistributionDevice**
- Renamed class **ThermalDistrubution** to **ThermalDistributionDevice**

Layered Construction package

- Class **MaterialLibrary**
 - Changed multiplicity of property *libraryMember* to 0..*
- Class **LayeredConstructionLibrary**
 - Changed multiplicity of property *libraryMember* to 0..*

Schedules package:

- Class **ScheduleLibrary**
 - Changed multiplicity of property *libraryMember* to 0..*

Cumulative list of changes since beta 7 (as of 2026-05-14)

CodeLists package:

Several CodeList classes added and some renamed. Added values to several CodeLists.

Enumerations package:

- Class **NetworkConnectionStatusValue** has been changed to a CodeList and moved to the corresponding package.

DataTypes package:

- Dropped class **Suitability**
- Added class **Absorptance**
- Added class **CityObjectRelation**
- Added class **FeatureRelation**
- Added class **GenericTimeValue**
- Added class **Indicator**
- Added class **Metadata**
- Added class **TimestampedValue**

Core package:

- Dropped class **AbstractBuildingUnit**
- Dropped class **AbstractUsageZone**
- Dropped class **AbstractThermalZone**
- Dropped class **RefurbishmentMeasure**

- Moved class **EnergyPerformanceCertification** to Building package
- Moved class **ADE _AbstractBuilding** to Building package
- Renamed class **AbstractBuildingPartition** to **AbstractCityObjectSpace**
- Renamed class **WeatherData** to **SensorData**
- Added class **AbstractADEFeature**
- Added class **AbstractFeatureWithLifeSpan**
- Added class **Intervention**
- Classes **AbstractResource**, **AbstractSchedule**, **AbstractLayeredConstruction**, **UtilityNetworkConnection**, **SensorData**, **DeviceOperation** are now derived from class **AbstractFeatureWithLifeSpan**
- Class **ADE _CityObject**:
 - Dropped property *suitability*
 - Renamed property from *weatherData* to *sensorData*
 - Added property *metadata*
 - Added property *identifier*
 - Added property *validFrom*
 - Added property *validTo*
 - Added property *status*
 - Added property *indicator*
- Class **AbstractLibrary**:
 - Dropped property *source*
 - Dropped property *author*
- Class **AbstractCityObjectSpace**:
 - Added property *lod0MultiSurface*
- Class **AbstractSchedule**:
 - Dropped property *libraryCode*
- Class **AbstractResource**:
 - Dropped property *status*
 - Dropped property *amountType*
 - Renamed property *costsMoney* to *expense*
 - Renamed property *yieldsMoney* to *revenue*
 - Added property *referencePeriod*
 - Added property *amountBasedOn*
 - Added property *timeDependentExpense*
 - Added property *timeDependentRevenue*
- Class **AbstractTimeSeries**:
 - Dropped property *acquisitionMethod*
 - Dropped property *source*
- Class **UtilityNetworkConnection**:
 - Dropped property *connectionStatus*

Building physics package:

Dropped. Some classes reworked and moved to new module Building

Occupancy package:

Dropped. Some classes reworked and moved to new module Building

Building package:

This module is new. It reworks and merges the previous Building physics module and Occupancy module. Additionally, it incorporates also class *ADE _AbstractBuilding* and its subclasses.

- Class **ADE _AbstractBuilding**
 - Class is moved from Core module
 - Dropped property *usageZone*
 - Dropped property *thermalZone*
 - Added property *occupiedBy*
 - Added property *zone*
 - Added property *owner*
 - Added property *ownershipType*
 - Added property *numberOfBuildingUnits*

- Class **EnergyPerformanceCertificate**
 - Class is moved from Core module
 - Class is derived from class **AbstractADEFeatureWithLifeSpan**
 - Dropped property *type*
 - Dropped property *issueDate*
 - Dropped property *expirationDate*
 - Dropped property *certificationUri*
- Class **Occupants**
 - Class is taken over from former Occupancy module
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
 - Renamed relation *occupancyRate* to *occupancySchedule*
- Added class **AbstractBuildingSpace**
- Class **BuildingUnit**
 - Class is taken over from former Occupancy module
 - Class is now derived from class **AbstractBuildingSpace**
 - Added property *floorNumberFrom*
 - Added property *floorNumberTo*
- Added class **AbstractZone**
- Added class **Zone**
- Added class **ZonePart**
- Class **ADE_BoundarySurface**
 - Class is taken over from former Building physics module
 - Dropped property *bdgBdrySurfIsAdiabatic*
 - Added property *bdgBdrySurfIsShared*
 - Added property *bdgBdryAdditionalThermalBridgeUValue*
- Added class **AbstractZoneBoundarySurface**
- Added class **ZoneAtticFloorSurface**
- Added class **ZoneBasementCeilingSurface**
- Added class **ZoneExteriorWallSurface**
- Added class **ZoneGroundSurfaceSurface**
- Added class **ZoneIntermediateFloorSurface**
- Added class **ZoneRoofSurface**
- Added class **ZoneUndergrounSurface**
- Added class **ZoneClosureSurface**
- Added class **ADE_Opening**
 - Class is taken over from former Building physics module
- Added class **ADE AbstractZoneOpening**
- Added class **ZoneDoor**
- Added class **ZoneWindow**

Devices package:

- Added class **GenericStorageDevice**
- Added class **EVChargingStation**
- Added class **ThermalDistribution**
- Added class **PowerDistribution**
- Class **AbstractDevice**
 - Added property *yearOfInstallation*

Urban function area package

- Class **UrbanFunctionArea**
 - Added property *area*
 - Set type of property *code* to gml:Code

WeatherStation package

- Added class **WeatherData**

Resources package

- Class **Energy**

- Added property *maximulLoad*
- Added property *maximulLoadTime*
- Added property *maximulLoadDay*
- Added property *maximulLoadMonth*

Layered Construction package

- Class **MaterialLibrary**
 - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionLibrary**
 - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionMaterial**
 - Dropped property *libraryCode*
 - Added property *type*
 - Added property *absorptance*
 - Added property *insideConvectionCoefficient*
 - Added property *outsideConvectionCoefficient*
- Class **AbstractMaterial**
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
 - Dropped property *libraryCode*
- Class **Layer**
 - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
- Class **SolidMaterial**
 - Added property *type*
 - Added property *isTransparent*
 - Added property *absorptance*
 - Added property *emissivity*
 - Added property *reflectance*
 - Added property *transmittance*
- Added class **Liquid**
- Class **Gas**
 - Added property *type*

Schedules package:

- Dropped class **AtomicSchedule**
- Dropped class **AtomicScheduleValue**
- Added class **AbstractAtomicSchedule**
- Added class **ConstantValueSchedule**
- Added class **TimeSeriesSchedule**
- Class **DualValueSchedule** is now derived from **AbstractAtomicSchedule**
- Class **ScheduleComponent** is now derived from **AbstractADEFeatureWithLifeSpan**

Time series package:

- Added class **AbstractTimeSeriesFile**
- Added class **MonthlyTimeSeries**
- Added class **MonthlyTimeSeriesFile**
- Added class **TypicalValuesMonthlyTimeSeries**
- Added class **TypicalValuesMonthlyTimeSeriesFile**
- Added class **IrregularTimeSeries**
- Added class **IrregularTimeSeriesFile**
- Added class **TypicalValuesIrregularTimeSeries**
- Added class **TypicalValuesIrregularTimeSeriesFile**
- Renamed class **TypicalValuesTimeSeries** to **TypicalValuesRegularTimeSeries**
- Renamed class **TypicalValuesTimeSeriesFile** to **TypicalValuesRegularTimeSeriesFile**
- Class **RegularTimeSeries**:
 - Dropped property *temporalExtent*
 - Added property *startTimestamp*
 - Added property *endTimestamp*
- Class **RegularTimeSeriesFile**:

- Class is now derived from class **AbstractTimeSeriesFile**
- Dropped property *temporalExtent*
- Dropped property *uom*
- Dropped property *fileURI*
- Dropped property *numberOfHeaderLines*
- Dropped property *fieldSeparator*
- Dropped property *recordSeparator*
- Dropped property *decimalSymbol*
- Dropped property *valueColumnNumber*
- Added property *startTimestamp*
- Added property *endTimestamp*
- Class **TypicalValuesRegularTimeSeriesFile**:
 - Class is now derived from class **AbstractTimeSeriesFile**
 - Dropped property *uom*
 - Dropped property *fileURI*
 - Dropped property *numberOfHeaderLines*
 - Dropped property *fieldSeparator*
 - Dropped property *recordSeparator*
 - Dropped property *decimalSymbol*
 - Dropped property *valueColumnNumber*

Version 3.0 beta 7

2025-10-14

No changes, except for version number going from 2.0 to 3.0.

Version 2.0 beta 7

2025-06-25

Schedules package:

- Class **ScheduleLibrary**
 - Changed relation type of *libraryMember* to composition

Layered Construction package:

- Class **LayeredConstructionLibrary**
 - Changed relation type of *libraryMember* to composition
- Class **LayeredConstruction**
 - Added property *gValue*
- Class **MaterialLibrary**
 - Changed relation type of *libraryMember* to composition

2025-05-30

Schedules package:

- Class **AtomicSchedule**
 - Changed multiplicity of relation *timeSeries* to [0..1]
 - Added property *constantValue*

2025-05-13

Building physics package:

- Class **ThermalZone**
 - Changed relation type of *thermalBoundary* to composition

2025-05-07**Core package:**

- Class **ADE_CityObject**
 - Changed relation type of *device* to aggregation
 - Changed relation type of *weatherData* to composition
- Class **AbstractThermalZone**
 - Changed relation type of *usageZone* to association
- Class **AbstractUsageZone**
 - Changed relation type of *buildingUnit* to association
- Class **AbstractResource**
 - Made relation to **AbstractTimeSeries** explicit and set to composition
- Class **DeviceOperation**
 - Made relation to **AbstractSchedule** explicit and set to aggregation
- Class **WeatherData**
 - Made relation to **AbstractTimeSeries** explicit and set to composition

Occupancy package:

- Class **UsageZone**
 - Changed relation type of *occupiedBy* to aggregation
 - Made all relations to **AbstractSchedule** explicit and set to aggregation
- Class **BuildingUnit**
 - Changed relation type of *occupiedBy* to aggregation
- Class **Occupants**
 - Made relation to **AbstractSchedule** explicit and set to aggregation

Layered Construction package:

- Class **LayeredConstruction**
 - Changed relation type of *layer* to composition
- Class **Layer**
 - Changed relation type of *material* to aggregation

Schedules package:

- Class **ScheduleComponent**
 - Changed relation type of *scheduleComponentMember* to aggregation

2025-05-01**Core package:**

- Class **ADE_AbstractSchedule**
 - Moved from package Schedules to Core
- Class **ADE_AbstractTimeSeries**
 - Moved from package Time Series to Core

Occupancy package:

- Class **Occupants:**
 - Added property *averageDietType*
 - Added property *averageIncomeLevel*
 - Added property *averageInstructionLevel*

CodeLists package:

- Added class **DietTypeValue**
- Added class **InstructionLevelValue**
- Added class **IncomeLevelValue**

2025-04-23**CodeLists package:**

- Added values to class **AreaTypeValue**

- Renamed class **HeightReferenceValue** to **HeightTypeValue**
- Added class **SuitabilityReasonValue**
- Added class **UrbanSpaceTypeValue**
- Added class **UrbanSpaceEndUseValue**

DataTypes package:

- Added class **Suitability**

Core package:

- Class **ADE_CityObject**
 - Added property *suitability*
- Class **ADE_AbstractBuilding**
 - Renamed property *bdgFloorArea* to *bdgArea*
 - Renamed property *bdgHeightAboveGround* to *bdgHeight*
- Class **AbstractBuildingPartition**
 - Renamed property *floorArea* to *area*

Resources package:

- Added class **UrbanSpace**

2025-04-12

Core package:

- Added class **AbstractLibrary**

Layered Construction package:

- Added class **LayeredConstructionLibrary**
- Added class **MaterialLibrary**

Schedules package:

- Added class **ScheduleLibrary**

CodeLists package:

- Class **FunctionInNetwork** renamed to **FunctionInNetworkValue**

Version 2.0 beta 6

2025-02-06

DataTypes package:

- Dropped class **HeatExchangeType**
- Dropped class **OpticalProperties**

Core package:

- Class **ADE_CityObject:**
 - Renamed property from *operationSchedule* to *deviceOperation*
- Class **AbstractDevice:**
 - Changed type of property *heatDissipation* to *gml:Measure*
 - Added property *heatDissipationConvectiveFraction*
 - Added property *heatDissipationLatentFraction*
 - Added property *heatDissipationRadiantFraction*

Occupancy package:

- Class **UsageZone:**
 - Renamed property *averageInternalGains* to *internalGains* and changed type to *gml:Measure*
 - Added property *internalGainsConvectiveFraction*
 - Added property *internalGainsLatentFraction*

- Added property *internalGainsRadiantFraction*
- Class **Occupants**:
 - Changed type of property *heatDissipation* to gml:Measure
 - Added property *heatDissipationConvectiveFraction*
 - Added property *heatDissipationLatentFraction*
 - Added property *heatDissipationRadiantFraction*

Layered Construction package:

- Class **LayeredConstruction**:
 - Dropped property *opticalProperties*
 - Added property *glazingRatio*
 - Added property *emissivity*
 - Added property *reflectance*
 - Added property *transmittance*
- Class **Layer**:
 - Changed multiplicity of property *thickness* to [1]
- Class **AbstractMaterial**:
 - Added property *libraryCode*

Version 2.0 beta 5

Resources package:

No changes since beta 4

Weather Station package:

No changes since beta 4

2025-01-30

Packages

- Created TimeSeries package and moved all related classes into it
- Created Schedule package and moved all related classes into it
- Dropped Supporting classes package

CodeLists package:

- Class **CertificateTypeValue** renamed to **EnergyPerformanceCertificateTypeValue**

2025-01-28

CodeLists package:

- Added class **ScheduleTypeValue**

Enumerations package:

- Dropped class **DayTypeValue**

Supporting classes package:

- Dropped class **TimeSeriesSchedule**
- Dropped class **DailyPatternSchedule**
- Dropped class **PeriodOfYear**
- Dropped class **DailySchedule**
- Added class **AtomicSchedule**
- Added class **CompositeSchedule**
- Added class **ScheduleComponent**
- Class **AbstractSchedule**:
 - Added properties *type*, *startTime*, *startDay*, *startMonth*, *startYear*, *temporalExtent*

2025-01-22

Devices package:

- Class **AbstractSolarCollector**:
 - Added properties *azimuth* and *inclination*
 - Dropped property *surfaceGeometry*
 - Added properties *lod2MultiSurface* and *lod3MultiSurface*

2025-01-13**CodeLists package:**

- Dropped class **ThermalBoundaryTypeValue**
- Added class **ShadingDeviceTypeValue**
- Added class **WeatherValueTypeValue**

DataTypes package:

- Moved class **PeriodOfYear** from package DataTypes to Supporting classes
- Moved class **DailySchedule** from package DataTypes to Supporting classes
- Class **AbstractQualifiedAttribute**:
 - Added properties *source* and *value*
- Classes **QualifiedHeight, QualifiedArea, QualifiedVolume**:
 - Dropped property *value*

Supporting classes package:

- Class **AbstractTimeSeries**:
 - Moved back from package Core to Supporting classes
 - Changed multiplicity of property *acquisitionMethod* to [1]
 - Changed multiplicity of property *interpolationType* to [1]
- Class **TypicalValuesTimeSeries**:
 - Added properties *startTime*, *startDay*, *startMonth*
- Class **TypicalValuesTimeSeriesFile**:
 - Added properties *startTime*, *startDay*, *startMonth*
- Added class **DualValueSchedule**:
- Class **PeriodOfYear**:
 - Changed stereotype from *dataType* to *featureType*
 - Added properties *startDay*, *startMonth*, *endDay*, *endMonth*
 - Dropped property *period*
- Class **DailySchedule**:
 - Changed stereotype from *dataType* to *featureType*
 - Added property *repetition*
 - Renamed property from *schedule* to *timeSeries*

Layered Construction package:

- Added class **ReverseLayeredConstruction**
- Dropped class **LayerComponent**
- Class **Layer**:
 - Removed property *layerComponent*
 - Added property *thickness*
 - Added property *material*
- Class **SolidMaterial**:
 - Renamed property from *specificHeat* to *specificHeatCapacity*
 - (Re)added and renamed property from *conductivity* to *thermalConductivity*
 - (Re)added property *embodiedEnergy*
 - (Re)added property *embodiedCarbon*
 - (Re)added property *porosity*

Core package:

- Class **AbstractDevice**:
 - Class is renamed from **AbstractEnergySystem** to **AbstractDevice**
 - Added property *heatDissipation*

- Added property *operationSchedule*
- Added property *efficiencyIndicator*
- Added property *installedPower*
- Added property *nominalEfficiency*
- Class **AbstractTimeSeries**:
 - Moved back from package Core to Supporting classes
- Class (ADE) **_CityObject**:
 - Renamed property from *energySystem* to *device*
- Class **OperationSchedule**:
 - Moved from package Devices to Core
- Class **WeatherData**:
 - Renamed property from *timeseries* to *timeDependentValues*
 - Renamed property from *singleValue* to *yearlyValue*
 - Renamed property from *singleValueType* to *valueType*
 - Changed type of property *valueType* from *CharacterString* to *CodeList*
- Added class **AbstractBuildingPartition**
- Classes **AbstractBuildingUnit**, **AbstractUsageZone**, **AbstractThermalZone**
 - Changed parent class

Building physics package:

- Dropped class **ThermalBoundary**
- Dropped class **ThermalOpening**
- Class **ThermalZone**
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
 - Removed property *boundedBy*
 - Added property *heatCapacity*
 - Added property *coincidesWithLod2Hull*
 - Added property *coincidesWithLod3Hull*
 - Added property *thermalBoundary*
- Added class **PartyWallSurface**
- Added class **BasementCeilingSurface**
- Added class **IntermediateFloorSurface**
- Added class **AtticFloorSurface**
- Class (ADE) **_BoundarySurface**:
 - Renamed property from *bdgBdrySrfWindowToWallRatio* to *bdgBdrySrfOpeningToSurfaceRatio*
 - Renamed property from *bdgBdrySrfArea* to *bdgBdryTotalSurfaceArea*
 - Added property *bdgBdryOpaqueSurfaceArea*
 - Added property *bdgBdrySrfThickness*
 - Added property *bdgBdrySrfHeatCapacity*

Occupancy package:

- Dropped class **AbstractFacilities**
- Dropped class **DHWFacilities**
- Dropped class **LightingFacilities**
- Dropped class **ElectricalAppliances**
- Dropped class **GenericFacilities**
- Class **BuildingUnit**:
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *equippedWith*
- Class **UsageZone**:
 - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
 - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
 - Removed property *equippedWith*

Devices package:

The former EnergySystems package is renamed to Devices package

- Class **OperationSchedule**:
 - Moved from package Devices to Core
- Dropped class **AbstractEnergyConversionSystem**
 - Now all subclasses are derived directly from **AbstractDevice**
- Dropped class **AbstractStorageSystem**
 - Now all subclasses are derived directly from **AbstractDevice**
- Class **GenericDevice**:
 - Renamed from class **GenericEnergyConversionSystem**
- Class **ElectricalStorageDevice**:
 - Renamed from class **PowerStorageSystem**
- Class **ThermalStorageDevice**:
 - Renamed from class **ThermalStorageSystem**
- Class **AbstractSolarCollector**:
 - Renamed from class **AbstractSolarEnergySystem**
- Class **PhotovoltaicThermalCollector**:
 - Renamed from class **PhotovoltaicThermalSystem**
- Class **PhotovoltaicCollector**:
 - Renamed from class **PhotovoltaicSystem**
- Class **SolarThermalCollector**:
 - Renamed from class **SolarThermalSystem**
- Class **GenericSolarCollector**:
 - Renamed from class **GenericSolarEnergySystem**
- Added class **LightingDevice**
- Added class **GenericElectricalDevice**
- Added class **MovableShadingDevice**

Urban Function Areas package:

- Class **UrbanFunctionArea**:
 - Removed property *rank*

Version 2.0 beta 4

2024-09-25

- Codelist **ThermalBoundaryTypeValue**
 - Renamed *outerWall* to *exteriorWall*
 - Renamed *intermediaryFloor* to *intermediateFloor*

2024-09-20

- Minor edits to the layout of the UML diagrams
- Polished Codelist **ThermalBoundaryTypeValue**

2024-08-12

- Class **ADE CityObject**:
 - Renamed association *connectedTo* to *utilityNetworkConnection*
 - Remodelled association class **CityObjectRelation**
- Class **AbstractTimeSeries**:
 - Moved to Core module
- Class **WeatherData**:
 - renamed property *valuesList* to *timeSeries*
 - made the association to **AbstractTimeSeries** explicit
- Added some values to codelist **RelationTypeValue**
- Class **TimeSeriesSchedule**:
 - Changed association to **AbstractTimeSeries** to Composition
- Class **DailySchedule**:
 - Changed association to **AbstractTimeSeries** to Composition
- Class **SensorConnection**:

- Changed type of *uom* to *UnitOfMeasure*
 - Renamed *linkToSensortDescription* to *linkToSensorDescription*
- Class **AbstractSolarEnergySystem**:
 - Removed association *installedOn* (now taken over by **CityObjectRelation**)
- Class **Energy**:
 - Changed multiplicity of *EnergyCarrier* to [0..1]
- Class **Waste**:
 - Changed multiplicity of *isDanegerous* and *isRecyclable* to [0..1]

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